

Physical AI Oncology Clinical Trial Site Complete Documentation Package

11 Documents for California's First Physical AI
Oncology Clinical Trial Site

Legislation, Regulations, Building Code,
Premises Code, and Operations

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SB 1042 - California Physical AI Oncology Clinical Trial Authorization and Site Establishment Act of 2026

1. Legislative Findings and Declarations

The Legislature finds and declares all of the following:

- (a) Physical artificial intelligence, including robotic surgical systems, collaborative robots, radio-therapy positioning systems, imaging assistants, rehabilitation exoskeletons, and companion robots, represents a transformative advancement in oncology clinical care delivery.

- (b) A 24-hour on-demand Physical AI oncology clinical trial simulation conducted on March 23, 2026, demonstrated that a single facility equipped with 10 robot types and 29 robot instances successfully served 168 unique patients across 15 cancer types in 24 hours, compared to 5 to 15 patients per day in traditional trial formats [1].
- (c) The simulation achieved a cumulative site Physical AI Standard Level (PSL) score of 63.4 to 64.4, within the Advanced Site band, with 99.7 percent facility uptime and an average patient wait time of 8 minutes.
- (d) All seven adverse events documented during the simulation were Grade 1 or Grade 2, all were resolved within the same hour they occurred, and zero patient harm resulted, demonstrating that Physical AI systems can manage both minute-level clinical details and sustained 24-hour operations [1].
- (e) The Physical AI trial format shifts control toward the patient by enabling patient-chosen scheduling on a 24/7 basis, increasing patient throughput by a factor of 10, reducing per-patient costs by 75 to 85 percent, and improving retention rates from 70 to 80 percent in traditional formats to 85 to 95 percent [2].
- (f) Staffing requirements are reduced from 80 to 120 full-time equivalents in traditional trial sites to 8 to 10 full-time equivalents in a Physical AI trial site, with the majority of clinical procedures, monitoring, documentation, and regulatory compliance tasks performed autonomously [3].
- (g) The Unification Standard Level (USL) framework provides a standardized 1.0 to 10.0 scoring system for evaluating physical AI robot readiness across four dimensions: simulation framework switching, AI integration, cross-robot progress sharing, and clinical trial collaboration [4].
- (h) Three adapted regulatory frameworks, covering ICH E6(R3) good clinical practice, 21 CFR Part 50 human subjects protection, and 21 CFR Part 312 investigational new drug applications, provide comprehensive governance for Physical AI oncology trials [5, 6, 7].
- (i) The cancer patient journey through a regulated Physical AI oncology trial has been documented across 10 sequential stages from pre-screening through data lock, demonstrating complete Response outcomes and a reduction in recurrence risk from 35 percent to 3 percent [8].
- (j) California has an opportunity to lead the nation in establishing safe, regulated, and patient-centric Physical AI oncology clinical trial sites that serve more patients with more robots and fewer workers while maintaining the highest standards of clinical care.

2. Definitions

For purposes of this chapter, the following definitions apply:

- (a) “Physical AI system” means a robotic or embodied artificial intelligence system that physically interacts with patients, clinical specimens, or medical equipment in the context of an oncology clinical trial. Physical AI systems include, but are not limited to, the following 10 categories:

- (1) Surgical robots that perform tumor resection through minimally invasive procedures.
 - (2) Collaborative robots (cobots) that stabilize patient limbs and assist with needle biopsy procedures.
 - (3) Radiotherapy patient-positioning robots that align patients with submillimeter precision for radiation delivery.
 - (4) Robotic needle-placement systems that use imaging guidance to place biopsy or ablation needles with high accuracy.
 - (5) Social companion robots that provide emotional support and anxiety reduction for patients, including pediatric patients.
 - (6) Humanoid robots that conduct rehabilitation exercises and patient engagement activities.
 - (7) Radiotherapy motion-management and tracking robots that monitor patient breathing and gate radiation beams accordingly.
 - (8) Imaging assistant robots that perform automated ultrasound and diagnostic imaging procedures.
 - (9) Steerable needle and needle-steering robots that guide flexible needles through tissue for targeted ablation.
 - (10) Rehabilitation exoskeletons and robotic gait trainers that support post-surgical mobility recovery.
- (b) “Physical AI Standard Level” or “PSL” means the scoring framework that evaluates each robot type on three regulatory dimensions: Omniscient (ICH E6(R3) compliance), Omnipresent (21 CFR Part 50 compliance), and Omnipotent (21 CFR Part 312 compliance), with each dimension scored from 0.0 to 10.0 per robot type.
- (c) “Unification Standard Level” or “USL” means the 1.0 to 10.0 scoring framework evaluating physical AI robot readiness across four equally weighted dimensions: simulation framework switching, AI integration, cross-robot progress sharing, and clinical trial collaboration [4].
- (d) “On-demand trial format” means a clinical trial structure in which patients may schedule procedures at any time during a 24-hour period, 7 days per week, with Physical AI systems providing continuous clinical care.
- (e) “Digital twin” means a patient-specific computational model calibrated to individual anatomy and treatment parameters, used for procedure planning, rehearsal, and intraoperative guidance.
- (f) “MCP-PAI standard” means the Model Context Protocol for Physical AI, a five-server interoperability topology providing authorization, FHIR, DICOM, ledger, and provenance services with hash-chained audit trails.
- (g) “Qualified site” means a facility that has met all building, premises, parking, staffing, equipment, and regulatory requirements established under this chapter and has received authorization from the department to conduct Physical AI oncology clinical trials.

- (h) “Department” means the California Department of Public Health.
- (i) “Sim-to-real validation” means the process of verifying that a Physical AI system’s performance in simulated environments transfers to real clinical settings within specified tolerances, including trajectory deviation below 2 millimeters and force discrepancy below 0.5 Newtons.

3. Authorization to Establish Physical AI Oncology Clinical Trial Sites

3.1. Site Authorization

- (a) The department shall establish a process for authorizing Physical AI oncology clinical trial sites in California, beginning with the first site in a prominent and safe San Francisco location.
- (b) A qualified entity may apply to the department for authorization to establish and operate a Physical AI oncology clinical trial site in a new building with dedicated parking facilities.
- (c) The authorization process shall include review of the following:
 - (1) Facility design and construction plans meeting the Physical AI oncology trial facility building code standards established under this chapter.
 - (2) Equipment specifications for all 10 Physical AI system categories to be deployed at the site.
 - (3) Staffing plans demonstrating adequate human oversight with a minimum of 8 to 10 full-time equivalent positions across three shifts.
 - (4) USL scores for each Physical AI system, with minimum thresholds of 7.0 for surgical systems, 6.0 for therapeutic positioning systems, 6.0 for diagnostic systems, 4.0 for rehabilitative systems, and 3.0 for companion systems.
 - (5) Phase 0 simulation validation results demonstrating cross-framework consistency using two or more simulation frameworks.
 - (6) Compliance plans for ICH E6(R3), 21 CFR Part 50, and 21 CFR Part 312 as adapted for Physical AI systems.
 - (7) Emergency response and cybersecurity plans.
 - (8) Parking and patient transportation accessibility plans.
- (d) The department shall act on a complete application within 120 days of receipt.

3.2. Statewide Applicability

- (a) The authorization requirements and standards established under this chapter shall apply uniformly to Physical AI oncology clinical trial sites throughout the State of California.

- (b) A city, county, or city and county may adopt additional requirements for Physical AI oncology clinical trial sites, provided those requirements do not conflict with or impose lesser standards than those established under this chapter.
- (c) The department shall publish model guidelines to assist local jurisdictions in integrating Physical AI oncology clinical trial site requirements into existing land use, building, and health facility permitting processes.

4. Physical AI System Requirements

4.1. USL and PSL Compliance

- (a) Each Physical AI system deployed at an authorized site shall maintain a USL score at or above the minimum threshold for its system category as specified in Section 3.1(c)(4).
- (b) The site shall maintain a cumulative PSL score that places it within the Advanced Site band (60.0 or above on the 0 to 100 scale) as a condition of continued authorization.
- (c) PSL scores shall be calculated and reported monthly using the three regulatory dimensions established in the PSL framework, with changes limited to a maximum of 0.3 per hour per dimension per robot type to ensure scoring stability [9].
- (d) A site whose cumulative PSL score falls below the Advanced Site band for two consecutive monthly reporting periods shall submit a corrective action plan to the department within 30 days.

4.2. Simulation Validation Requirements

- (a) Before deployment at a qualified site, each Physical AI system shall complete Phase 0 simulation validation using two or more validated frameworks, which may include NVIDIA Isaac Lab, MuJoCo, Gazebo Sim, and PyBullet [5].
- (b) Cross-framework validation shall demonstrate trajectory deviation below 2 millimeters and force discrepancy below 0.5 Newtons.
- (c) Digital twin models shall achieve a minimum fidelity of 90 percent when compared to patient-specific imaging and anatomical data.
- (d) Failure mode analysis shall be conducted across a minimum of 1,000 simulated procedures per Physical AI system type before clinical deployment [7].

4.3. Informed Consent and Patient Rights

- (a) Physical AI oncology clinical trial sites shall comply with all informed consent requirements established in the adapted 21 CFR Part 50 for Physical AI systems, including disclosure of robotic system roles, USL scores in plain language, safety records, and emergency stop provisions [6].

- (b) Patients shall have the right to schedule procedures at any time during the 24-hour operating period, consistent with the on-demand trial format.
- (c) Patients shall have the right to request human intervention or oversight at any point during a Physical AI-assisted procedure.
- (d) Pediatric patients shall receive age-appropriate robotic demonstrations and assent procedures as specified in the adapted 21 CFR Part 50 § 50.50 through § 50.56 [6].
- (e) Each patient shall be informed of the specific robots that will be involved in their care, the role of each robot (autonomous, assistive, or shared-autonomy), and the qualifications of human oversight personnel [10].

5. Safety and Reporting Requirements

5.1. Adverse Event Reporting

- (a) The site shall report all Physical AI-related adverse events to the department and to the Food and Drug Administration in accordance with the timelines established in the adapted 21 CFR Part 312 § 312.32, including 7-day reporting for fatal or life-threatening events and 15-day reporting for serious events [7].
- (b) Seven categories of reportable Physical AI adverse events shall be tracked: serious Physical AI adverse events, Physical AI-drug interaction events, cybersecurity incidents, AI model performance degradation, sim-to-real divergence, digital twin discrepancy, and safety review committee findings.
- (c) The site shall maintain a Physical AI Safety Review Committee with expertise in robotic engineering, AI and machine learning, oncology, and cybersecurity, meeting at intervals not exceeding 90 days.

5.2. Data Protection and Audit Trails

- (a) All Physical AI systems shall maintain hash-chained audit trails using SHA-256 or equivalent cryptographic standards, as required by the adapted ICH E6(R3) and 21 CFR Part 11 [5].
- (b) Patient data processed through MCP-PAI servers shall be de-identified using HIPAA Safe Harbor methodology with deny-by-default authorization policies.
- (c) The site shall implement federated learning protocols with differential privacy protections for multi-site data sharing, as validated in the simulation where 70 federated learning rounds across 12 sites used secure multi-party computation with differential privacy at epsilon equals 1.0 [8].

6. Facility and Infrastructure Standards

- (a) Physical AI oncology clinical trial sites shall be constructed in accordance with the building code standards established under this chapter, including a minimum floor area of 85,000 to 100,000 square feet, HEPA filtration in surgical suites, shielded radiotherapy vaults, and dedicated server rooms for AI computation infrastructure [3].
- (b) The site shall provide dedicated parking facilities meeting the standards established under this chapter, including accessible parking, patient drop-off zones, and emergency vehicle access.
- (c) Power infrastructure shall support 800 to 1,200 kilowatts of sustained electrical load with redundant power supply and uninterruptible power systems for all critical clinical areas.
- (d) Network infrastructure shall provide a minimum of 10 gigabits per second internal bandwidth with cybersecurity protections meeting the requirements of the MCP-PAI standard [3].

7. Workforce Transition and Training

- (a) The department, in consultation with the Employment Development Department and the California Workforce Development Board, shall establish programs to assist clinical trial workers in transitioning to roles in Physical AI trial sites or related fields.
- (b) Each authorized site shall provide comprehensive training for all human staff in Physical AI system operation, emergency response, patient interaction, and regulatory compliance.
- (c) Training requirements shall include demonstrated competency in understanding Physical AI system capabilities, limitations, and failure modes, as well as the ability to intervene manually when necessary.
- (d) The Legislature recognizes that while Physical AI trial sites require fewer workers (8 to 10 full-time equivalents compared to 80 to 120 in traditional formats), the transition enables more patients to be treated, with the simulation demonstrating 168 patients served in 24 hours compared to approximately 5 to 15 per day traditionally.

8. Relationship to Existing California Law

This chapter is intended to complement and operate in conjunction with existing California laws governing health care and artificial intelligence, including but not limited to the following:

- (a) AB 489 (2025), concerning health care professions and deceptive terms related to artificial intelligence, which ensures clear communication about AI involvement in clinical care.
- (b) AB 3030 (2024), concerning health care services and artificial intelligence, which establishes transparency requirements for AI in health care delivery.
- (c) SB 1120 (2024), concerning health care coverage and utilization review, which addresses AI-assisted clinical decision-making.

- (d) SB 243 (2025), concerning companion chatbots, which establishes safety standards for AI companion systems relevant to social companion robots used in Physical AI oncology trials.
- (e) AB 2013 (2024), concerning generative artificial intelligence and training data transparency, which applies to the generative AI and VLA models integrated into Physical AI systems.

9. Appropriation

The sum of fifty million dollars (\$50,000,000) is hereby appropriated from the General Fund to the department for the purposes of implementing this chapter, including the establishment of the site authorization process, development of model guidelines, workforce transition programs, and ongoing oversight of authorized Physical AI oncology clinical trial sites.

10. Effective Date

This act shall take effect on January 1, 2027.

AB 2847 - California Physical AI Patient Rights and Robotic Safety Act of 2026

1. Legislative Findings and Declarations

The Legislature finds and declares all of the following:

- (a) Physical AI oncology clinical trials represent a fundamental shift in the relationship between patients and clinical care systems, in which control has now shifted towards the patient's side, with more patient rights than traditional clinical trial formats provide.
- (b) The 24-hour on-demand Physical AI oncology trial simulation demonstrated that patients can choose their own scheduling across a full 24-hour period, experience average wait times of 8 minutes, and achieve patient satisfaction scores of 4.7 out of 5.0, compared to the rigid scheduling and extended wait times typical of traditional oncology trials [1].
- (c) More patients will be treated with more robots and fewer workers, with the simulation demonstrating 168 patients served by 29 robot instances staffed by 8 to 10 human full-time equivalents across three shifts, compared to 5 to 15 patients per day in traditional formats requiring 80 to 120 staff [2].
- (d) The cancer patient journey through a regulated Physical AI oncology trial documented a complete patient trajectory from pre-screening through data lock across 10 stages, achieving complete Response, R0 resection, and reduction in recurrence risk from 35 percent to 3 percent, demonstrating that patient outcomes improve under Physical AI care [8].

- (e) Pediatric patients require special protections, including age-appropriate demonstrations of robotic systems, parental consent provisions, and dedicated companion robot interactions, as established in the adapted 21 CFR Part 50 § 50.50 through § 50.56 [6].
- (f) Patient-facing robot instructions for all 10 robot types have been developed and validated, providing step-by-step guidance for patients interacting with surgical robots, cobots, radiotherapy positioning systems, needle-placement systems, social companions, humanoids, motion-tracking systems, imaging assistants, steerable needles, and rehabilitation exoskeletons [10].
- (g) The simulation demonstrated that both specific details by the minute and details over 24 hours were handled correctly by AI, with minute-level vital sign monitoring, real-time adverse event detection and resolution, and sustained clinical operations across all 24 hours.
- (h) Seven adverse events occurred across 168 patients (4.2 percent incidence rate), all Grade 1 or Grade 2, all resolved within the same hour, demonstrating a safety profile that supports expanding patient access to Physical AI oncology care [1].

2. Patient Bill of Rights for Physical AI Oncology Clinical Trials

2.1. Right to On-Demand Scheduling

- (a) Every patient enrolled in a Physical AI oncology clinical trial at an authorized California site shall have the right to schedule procedures, consultations, and follow-up visits at any time during the site's 24-hour operating period, 7 days per week, subject only to clinical safety constraints and equipment availability.
- (b) The site shall provide scheduling systems that enable patients to select appointment times without requiring approval from clinical staff, with the Physical AI systems managing patient flow and resource allocation autonomously.
- (c) Maximum wait times from arrival to procedure initiation shall not exceed 30 minutes under normal operating conditions. The simulation demonstrated an average wait time of 8 minutes across 168 patients [1].

2.2. Right to Full Disclosure

- (a) Before any Physical AI-assisted procedure, the patient shall receive clear and complete disclosure of the following:
 - (1) The specific Physical AI system or systems that will be involved in their care, identified by type, model, and the role each system will perform (primary, secondary, or ancillary).
 - (2) Whether the system will operate in autonomous, assistive, or shared-autonomy mode.
 - (3) The system's current USL score presented in plain language, with an explanation of what the score means for the patient's procedure [4].

- (4) The system’s safety record, including any prior adverse events involving the specific system instance.
 - (5) Emergency stop provisions and how they will be activated if needed.
 - (6) The identity and qualifications of the human oversight personnel responsible for supervising the procedure.
 - (7) Simulation validation results relevant to the patient’s specific procedure type.
 - (8) Digital twin usage and the patient’s right to request deletion of their digital twin data at any time.
- (b) Disclosure materials shall be available in the patient’s preferred language and in formats accessible to patients with visual, auditory, or cognitive impairments.
 - (c) The site shall provide patient-facing robot instructions consistent with those developed for each of the 10 Physical AI system categories [10].

2.3. Right to Human Intervention

- (a) A patient shall have the right to request human intervention or oversight at any point during a Physical AI-assisted procedure, and the site shall be staffed to accommodate such requests within 2 minutes of the request.
- (b) A patient may withdraw consent for Physical AI involvement at any time and request that the remainder of their procedure be completed by human clinicians, to the extent that such transition is clinically safe.
- (c) The site shall maintain at least one on-call emergency physician and one on-call pharmacist at all times during the 24-hour operating period, in addition to the site safety officers present on each shift.
- (d) Emergency stop mechanisms shall be accessible to both clinical staff and patients where clinically appropriate, with clear instructions provided in advance.

2.4. Right to Privacy and Data Control

- (a) All patient data collected, processed, or stored by Physical AI systems shall be protected in accordance with HIPAA, the California Consumer Privacy Act, the California Privacy Rights Act, and the data governance requirements of the adapted ICH E6(R3) [5].
- (b) Patient data processed through MCP-PAI servers shall use deny-by-default authorization policies, meaning no data access is permitted unless explicitly authorized for a specific purpose [5].
- (c) Patients shall have the right to request a complete record of all Physical AI system interactions involving their care, including robotic telemetry logs, AI decision logs, and digital twin state data.

- (d) Patients shall have the right to request deletion of their digital twin and associated patient-specific data from Physical AI system local storage, subject to regulatory data retention requirements under 21 CFR Part 312.
- (e) Hash-chained audit trails (SHA-256) shall be maintained for all patient data transactions, providing an immutable record of data access and modification [5].

2.5. Right to Equitable Access

- (a) Physical AI oncology clinical trial sites shall not discriminate in enrollment, scheduling, or treatment on the basis of race, ethnicity, sex, gender identity, sexual orientation, age, disability, socioeconomic status, insurance status, or geographic location.
- (b) The on-demand 24/7 scheduling format shall accommodate patients who work non-traditional hours, have caregiving responsibilities, or face transportation barriers, consistent with the patient-centric design demonstrated in the simulation [2].
- (c) The site shall provide transportation assistance, including accessible parking, patient drop-off zones, and coordination with public transit and ride-share services.
- (d) The per-patient cost reduction of 75 to 85 percent demonstrated in the simulation format comparison shall be reflected in reduced financial barriers for patients [2].

2.6. Pediatric Patient Protections

- (a) Pediatric patients (under 18 years of age) shall receive age-appropriate explanations and demonstrations of all Physical AI systems involved in their care, consistent with the assent requirements of the adapted 21 CFR Part 50 § 50.50 through § 50.56 [6].
- (b) Social companion robots used with pediatric patients shall follow the interaction protocols established in the patient robot instructions, including gentle high-fives, interactive games, calm-breathing prompts, and virtual sticker rewards [10].
- (c) Humanoid robots conducting rehabilitation exercises with pediatric patients shall follow established protocols including slow count-in and count-out breathing, posture cues, and interaction durations appropriate for the child's age and condition [10].
- (d) A parent or guardian shall be present or immediately available during all Physical AI-assisted procedures involving pediatric patients, and an advocate shall be appointed for wards of the state consistent with § 50.56 of the adapted regulations [6].
- (e) The simulation demonstrated successful management of pediatric patients including pediatric ALL (acute lymphoblastic leukemia) and AML (acute myeloid leukemia) cases, with companion robot-assisted anxiety reduction achieving 95 percent effectiveness [1].

3. Robotic Safety Standards

3.1. Physical AI System Safety Certification

- (a) Before deployment at an authorized California site, each Physical AI system shall obtain safety certification through a process administered by the department, demonstrating the following:
 - (1) Completion of Phase 0 simulation validation across two or more validated simulation frameworks with cross-framework trajectory deviation below 2 millimeters and force discrepancy below 0.5 Newtons [5].
 - (2) Achievement of minimum USL scores: 7.0 for surgical robots, 6.0 for therapeutic positioning and diagnostic systems, 4.0 for rehabilitative systems, and 3.0 for companion systems [6].
 - (3) Completion of failure mode analysis across a minimum of 1,000 simulated procedures [7].
 - (4) Functional emergency stop mechanisms tested and verified.
 - (5) Force-limiting capabilities appropriate to the system type, including maximum 250 millimeters per second approach speed and force-limited contact for cobots [10].
- (b) Safety certification shall be renewed annually and following any significant hardware or software modification.

3.2. Continuous Safety Monitoring

- (a) Each Physical AI system shall maintain continuous telemetry recording during all patient interactions, as demonstrated in the hourly robot logs of the 24-hour simulation where all 29 robot instances were tracked with active and standby status, performance metrics, and state transitions [1].
- (b) Real-time safety monitoring shall include automated detection of force limit exceedances, positioning errors beyond specified tolerances, AI model performance degradation, and cybersecurity anomalies.
- (c) The simulation demonstrated 99.7 percent fleet uptime across 29 robot instances over 24 hours, with one planned maintenance event (SURG-01 maintenance window from 22:15 to 01:59) managed without patient impact [1].
- (d) Sim-to-real divergence exceeding 4 millimeters in trajectory or 1.0 Newton in force shall trigger an automatic safety review and potential suspension of the affected system [7].

3.3. Adverse Event Response

- (a) Every Physical AI-related adverse event shall be documented, reported, and resolved in accordance with the timelines and categories established in the adapted 21 CFR Part 312 § 312.32 [7].

- (b) The site shall achieve a target of resolving all Grade 1 and Grade 2 adverse events within the same hour of occurrence, consistent with the performance demonstrated in the simulation where all seven adverse events (ranging from nausea and hypotension to pain, cough, desaturation, and anxiety) were resolved within the hour [1].
- (c) A Physical AI Safety Review Committee shall convene within 24 hours of any Grade 3 or higher adverse event and within 7 days of any fatal or life-threatening event.
- (d) The committee shall include expertise in robotic engineering, AI and machine learning, oncology, and cybersecurity, meeting at regular intervals not exceeding 90 days [7].

4. Patient Complaint and Grievance Process

- (a) Each authorized site shall establish a patient complaint and grievance process that is accessible 24 hours per day, 7 days per week, consistent with the site's operating schedule.
- (b) Complaints involving Physical AI system malfunctions or safety concerns shall be investigated within 48 hours and the patient shall be notified of the investigation outcome within 14 days.
- (c) The department shall maintain a public registry of Physical AI-related patient complaints and their resolutions, with appropriate de-identification of patient information.
- (d) Patients shall have the right to file complaints directly with the department if they believe the site's internal grievance process has not adequately addressed their concerns.

5. Enforcement

- (a) The department shall have authority to inspect authorized Physical AI oncology clinical trial sites without prior notice to verify compliance with this article.
- (b) Violations of patient rights established under this article shall be subject to civil penalties of not less than one thousand dollars (\$1,000) and not more than twenty-five thousand dollars (\$25,000) per violation.
- (c) The department may suspend or revoke site authorization for repeated or serious violations of patient rights or robotic safety standards.
- (d) Nothing in this article shall be construed to limit any other remedy available to patients under California law.

6. Effective Date

This act shall take effect on January 1, 2027.

SB 892 - California Physical AI Clinical Data Protection and Transparency Act of 2026

1. Legislative Findings and Declarations

The Legislature finds and declares all of the following:

- (a) Physical AI oncology clinical trials generate unprecedented volumes of patient data, including robotic telemetry, AI decision logs, digital twin state data, medical imaging, vital signs captured at minute-level resolution, and federated learning model parameters [1].
- (b) The 24-hour on-demand simulation generated 178 output files across 25 sequential commits, including hourly patient records with minute-level vital sign data for 168 patients, detailed robot telemetry for 29 instances across 24 hours, and PSL scoring data for all 10 robot types [1].
- (c) The MCP-PAI standard establishes a five-server interoperability topology (authorization, FHIR, DICOM, ledger, provenance) with 23 tools, six actor roles, five conformance levels, and hash-chained audit trails, providing a comprehensive data governance architecture [5].
- (d) Training data transparency requirements established by AB 2013 (2024) apply to the generative AI and VLA models integrated into Physical AI systems, and additional transparency requirements specific to robotic telemetry and clinical decision data are needed.
- (e) The cancer patient journey demonstrated federated learning across 12 sites using secure multi-party computation with differential privacy at epsilon equals 1.0, providing a model for privacy-preserving multi-site data sharing [8].
- (f) Both specific details by the minute and details over 24 hours were captured and managed correctly by AI systems, creating a data governance challenge that requires clear rules for data retention, access, sharing, and deletion across multiple timescales.

2. Definitions

For purposes of this chapter, the following definitions apply:

- (a) “Physical AI clinical data” means any data generated, collected, processed, or stored by Physical AI systems during oncology clinical trials, including robotic telemetry data, AI decision logs, digital twin state data, patient vital signs, medical imaging data, procedural records, adverse event records, and PSL and USL scoring data.
- (b) “Robotic telemetry data” means real-time operational data from Physical AI systems including position, force, velocity, temperature, power consumption, active or standby status, and performance metrics.

- (c) “AI decision log” means a record of decisions made by AI algorithms integrated into Physical AI systems, including treatment recommendations, scheduling decisions, resource allocation, and safety interventions.
- (d) “Digital twin data” means patient-specific computational model data including anatomical models, treatment simulation parameters, procedure rehearsal records, and predictive model outputs.
- (e) “Hash-chained audit trail” means an immutable, cryptographically linked sequence of data access and modification records using SHA-256 or equivalent algorithms, as specified in the adapted ICH E6(R3) [5].
- (f) “Federated learning data” means model parameters, gradient updates, and aggregated statistics shared between Physical AI systems across multiple clinical trial sites using privacy-preserving computation methods.
- (g) “De-identification” means the process of removing or obscuring personal health information using the HIPAA Safe Harbor methodology with 18 identifier categories, as required by the adapted regulatory frameworks [6].

3. Data Collection Transparency

3.1. Notice Requirements

- (a) Before enrollment, each patient shall receive a comprehensive data collection notice specifying all categories of Physical AI clinical data that will be generated during their participation, the purposes for which each category will be used, and the retention period for each category.
- (b) The notice shall identify the five MCP-PAI servers that will process patient data (authorization, FHIR, DICOM, ledger, provenance) and describe the role of each server in plain language [5].
- (c) The notice shall disclose whether patient data will be used for federated learning and, if so, describe the differential privacy protections applied, including the privacy budget (epsilon value) [8].
- (d) The notice shall be updated whenever new categories of data collection are introduced or existing categories are materially changed.

3.2. Real-Time Data Access

- (a) Patients shall have the right to access their Physical AI clinical data in real time through a secure patient portal, including current vital signs, procedure status, robot telemetry relevant to their care, and digital twin visualizations.
- (b) The site shall provide data access in standardized formats, including FHIR for clinical data and DICOM for medical imaging data.

- (c) Data access requests shall be fulfilled within 24 hours for historical data not available through the real-time portal.

4. Data Protection Standards

4.1. Encryption and Access Control

- (a) All Physical AI clinical data shall be encrypted at rest using AES-256 or equivalent and in transit using TLS 1.3 or later.
- (b) Access to Physical AI clinical data shall follow deny-by-default authorization policies, meaning no access is permitted unless explicitly authorized for a specific purpose, role, and time period [5].
- (c) The six MCP-PAI actor roles (patient, clinician, investigator, sponsor, regulator, system) shall define the maximum data access scope for each role, with no role having unrestricted access to all data categories [5].
- (d) Multi-factor authentication shall be required for all human access to Physical AI clinical data systems.

4.2. Audit Trail Requirements

- (a) Hash-chained audit trails shall be maintained for all Physical AI clinical data transactions, providing cryptographic proof of data integrity and an immutable record of who accessed what data, when, and for what purpose [5].
- (b) Audit trail records shall be retained for a minimum of 15 years following the completion of the clinical trial, or longer if required by applicable regulations.
- (c) The site shall conduct quarterly audit trail integrity verification using cryptographic chain validation.
- (d) Audit trail data shall be stored in a separate system from clinical data to prevent simultaneous compromise.

4.3. De-identification Standards

- (a) All Physical AI clinical data shared outside the treating site shall be de-identified using the HIPAA Safe Harbor methodology, removing all 18 categories of identifiers [6].
- (b) Robotic telemetry data shall be assessed for re-identification risk before sharing, as movement patterns and procedural signatures may constitute quasi-identifiers.
- (c) Digital twin data shall be fully de-identified before any use in federated learning, research, or quality improvement activities.

- (d) The site shall employ a qualified statistician or privacy officer to verify that de-identified data sets meet the threshold for non-identifiability.

5. Federated Learning and Multi-Site Data Sharing

- (a) Physical AI oncology clinical trial sites that participate in federated learning shall implement the privacy-preserving computation methods demonstrated in the cancer patient journey, including secure multi-party computation with differential privacy [8].
- (b) The privacy budget (epsilon value) for differential privacy shall not exceed 1.0 per federated learning round without explicit patient consent and IRB approval.
- (c) Model parameters shared through federated learning shall be verified to not contain memorized patient data through membership inference testing.
- (d) Federated learning participants shall maintain data harmonization standards using DICOM, FHIR, ICD-10 to SNOMED CT mapping, and LOINC coding [5].
- (e) The simulation demonstrated that 70 federated learning rounds across 12 sites achieved clinically significant model improvements while maintaining patient privacy, providing evidence that multi-site Physical AI data sharing can be conducted safely [8].

6. Data Retention and Deletion

- (a) Physical AI clinical data shall be retained for the duration required by applicable federal and state regulations, including a minimum of 2 years following investigational drug approval or 2 years after investigation discontinuation, consistent with 21 CFR Part 312 [7].
- (b) Upon trial completion and satisfaction of regulatory retention requirements, patients shall have the right to request deletion of their digital twin data and patient-specific data from Physical AI system local storage.
- (c) Deletion requests shall be fulfilled within 30 days using secure data erasure methods that comply with HIPAA Safe Harbor requirements, and the site shall provide written confirmation of deletion [7].
- (d) Aggregated, de-identified data may be retained indefinitely for research purposes, provided it meets the de-identification standards established in Section 4.3.
- (e) When a site's authorization is revoked or a trial is terminated, all patient-specific data shall be securely erased from Physical AI system local storage within 30 days of decommissioning, consistent with the IND withdrawal requirements of the adapted 21 CFR Part 312 § 312.38 [7].

7. Cybersecurity Incident Reporting

- (a) Any cybersecurity incident affecting Physical AI clinical data shall be reported to the department, to affected patients, and to the FDA in accordance with the following timelines:
 - (1) Within 7 days if the incident involves potential patient harm or breach of protected health information for 500 or more individuals.
 - (2) Within 15 days for all other cybersecurity incidents affecting Physical AI systems [7].
- (b) The site shall maintain a cybersecurity incident response plan that addresses Physical AI-specific threats including robot control system compromise, AI model poisoning, digital twin data manipulation, and MCP-PAI server infrastructure attacks.
- (c) Annual cybersecurity assessments shall include penetration testing of Physical AI system interfaces, MCP-PAI server infrastructure, and network segmentation between clinical and administrative systems.
- (d) The simulation demonstrated continuous operation across 24 hours with no cybersecurity incidents, including secure management of 10 gigabits per second internal network traffic supporting 29 robot instances and patient data systems [3].

8. AI Model Transparency

- (a) In accordance with the principles of AB 2013 (2024), sites shall maintain and make available upon request documentation of the AI models integrated into Physical AI systems, including model architecture, training data sources, validation methodology, and performance benchmarks.
- (b) The five categories of AI recognized in the adapted ICH E6(R3) - generative AI (VLA models, diffusion policies), agentic AI (LLM-based assistants), reinforcement learning, self-supervised learning, and supervised learning - shall each have category-specific transparency requirements [5].
- (c) Pre-determined change control plans (PCCPs) for AI model updates shall be filed with the department and made available to patients upon request, specifying which model changes require protocol amendments versus those within the PCCP scope [7].
- (d) AI model performance shall be reported in the site's annual report alongside PSL and USL scores, providing a comprehensive transparency record [7].

9. Enforcement and Penalties

- (a) Violations of data protection standards established under this chapter shall be subject to civil penalties of not less than two thousand five hundred dollars (\$2,500) and not more than seven thousand five hundred dollars (\$7,500) per violation per patient affected.

- (b) Intentional or reckless failure to maintain hash-chained audit trails or to report cybersecurity incidents within required timelines shall constitute an aggravated violation subject to penalties of up to twenty-five thousand dollars (\$25,000) per violation.
- (c) The Attorney General may bring an action to enforce this chapter on behalf of the people of the state.
- (d) Nothing in this chapter shall be construed to limit any private right of action available to patients under the California Consumer Privacy Act or the California Privacy Rights Act.

10. Effective Date

This act shall take effect on January 1, 2027.

San Francisco Municipal Code Update - Physical AI Oncology Clinical Trial Site Requirements

1. Purpose and Scope

1.1. Purpose

This regulation establishes the municipal permit, zoning, and operational requirements for Physical AI oncology clinical trial sites within the City and County of San Francisco, in furtherance of the California Physical AI Oncology Clinical Trial Authorization and Site Establishment Act (SB 1042). San Francisco shall be the location of the first Physical AI oncology clinical trial site in California, situated in a prominent and safe location that provides accessible care to oncology patients throughout the city and surrounding Bay Area communities.

1.2. Scope

These requirements apply to the permitting, construction, and operation of new-construction facilities specifically designed as Physical AI oncology clinical trial sites, including the associated parking facilities required under this regulation. These requirements supplement but do not replace existing San Francisco Planning Code, Building Code, and Health Code requirements for medical facilities.

1.3. Evidence Base

The 24-hour on-demand Physical AI oncology clinical trial simulation conducted on March 23, 2026, provides the primary evidence base for these regulations, demonstrating that a single 85,000 to 100,000 square-foot facility equipped with 10 robot types (29 total instances) served 168 unique

patients across 15 cancer types with 99.7 percent facility uptime and zero patient harm [1]. The simulation demonstrated that both specific clinical details at minute-level resolution and sustained operations across a full 24-hour cycle were managed correctly by Physical AI systems [1].

2. Zoning Requirements

2.1. Permitted Zones

- (a) Physical AI oncology clinical trial sites shall be a conditionally permitted use in the following San Francisco zoning districts:
 - (1) Neighborhood Commercial (NC) districts with parcels of 20,000 square feet or larger.
 - (2) Mixed Use-General (MUG) districts.
 - (3) Production, Distribution, and Repair (PDR) districts adjacent to residential or commercial zones with transit access.
 - (4) Downtown General (DTG) and Downtown Office (C-3) districts.
 - (5) Medical Use Districts and Health Care Services Districts.
- (b) Preference shall be given to locations within one-quarter mile of a Bay Area Rapid Transit (BART) station or San Francisco Municipal Railway (Muni) stop to maximize patient accessibility consistent with the on-demand 24/7 scheduling model [2].
- (c) The site shall be located in a prominent and safe location, defined as a parcel with frontage on a street classified as an arterial or collector in the San Francisco General Plan, with adequate street lighting, pedestrian infrastructure, and proximity to emergency services.

2.2. Setback and Height Requirements

- (a) The minimum lot size for a Physical AI oncology clinical trial site shall be 40,000 square feet, accommodating the 85,000 to 100,000 square-foot building footprint across multiple stories plus dedicated parking facilities [3].
- (b) Building height shall comply with the applicable district height limit but shall not exceed 65 feet unless a variance is obtained through the Planning Commission conditional use process.
- (c) Street-level setbacks shall provide a minimum 15-foot pedestrian zone along all public street frontages.
- (d) The site shall include a dedicated patient drop-off and pick-up zone accessible from the public right-of-way, with a covered canopy extending a minimum of 20 feet from the building entrance.

3. Conditional Use Permit Requirements

3.1. Application Contents

An application for a conditional use permit for a Physical AI oncology clinical trial site shall include, in addition to standard Planning Code requirements, the following:

- (a) A Physical AI System Deployment Plan identifying all 10 robot types to be deployed, the number of instances of each type, and the USL and PSL scores for each system, demonstrating compliance with the minimum thresholds established by state law [4, 9].
- (b) A Patient Flow Analysis based on the simulation data, including projected daily patient volumes (up to 168 patients per 24 hours), arrival and departure patterns, peak utilization periods (the simulation showed a peak of 15 arrivals in hour 09 with 72.4 percent utilization and 28 concurrent patients on-site), and measures to manage queuing and wait times [1].
- (c) A Neighborhood Impact Assessment addressing noise, traffic, parking demand, emergency vehicle access, and the visual impact of the facility on the surrounding neighborhood.
- (d) A Community Benefits Plan describing workforce development programs, local hiring commitments, and patient access programs for San Francisco residents.
- (e) A Transportation Demand Management (TDM) Plan meeting the requirements of San Francisco Planning Code Section 169, reflecting the 24/7 operating schedule and the reduced staffing model (8 to 10 full-time equivalents versus 80 to 120 in traditional trial sites) [3].

3.2. Public Notice and Hearing

- (a) The Planning Department shall provide notice of the conditional use permit application to all owners and occupants of property within 300 feet of the proposed site.
- (b) A public hearing before the Planning Commission shall be held not fewer than 30 days after notice is provided.
- (c) The applicant shall hold at least one community meeting in the neighborhood where the site is proposed, at least 14 days before the Planning Commission hearing.

4. Building Permit Requirements

4.1. Structural and Mechanical Systems

- (a) Building permit applications for Physical AI oncology clinical trial sites shall demonstrate compliance with the California Building Standards Code (Title 24) as supplemented by the San Francisco Building Code, with the following additional requirements specific to Physical AI operations:

- (b) Surgical suites shall incorporate HEPA filtration (ISO Class 7 or better), positive-pressure air handling, and floor load capacity of at least 150 pounds per square foot to support surgical robot systems and associated equipment [3].
- (c) Radiotherapy vaults shall include radiation shielding meeting the requirements of the California Radiologic Health Branch, with additional structural reinforcement to support the weight of radiotherapy positioning robot systems [3].
- (d) Server rooms for AI computation infrastructure shall be located on a separate HVAC zone with redundant cooling capable of rejecting a minimum heat load of 200 kilowatts.
- (e) The building shall accommodate 800 to 1,200 kilowatts of sustained electrical load with redundant utility feeds, automatic transfer switches, and uninterruptible power systems covering all critical clinical and data systems [3].

4.2. Accessibility

- (a) The site shall exceed the minimum accessibility requirements of the Americans with Disabilities Act and California Building Code Chapter 11B to ensure that the on-demand patient scheduling model is fully accessible to patients with mobility, visual, auditory, or cognitive impairments.
- (b) All patient circulation routes shall be designed to accommodate hospital beds, wheelchairs, and rehabilitation exoskeletons simultaneously.
- (c) Wayfinding systems shall include visual, auditory, and tactile elements appropriate for a facility where patients interact with 10 categories of robotic systems [10].

5. Health Code Requirements

5.1. Operating License

- (a) A Physical AI oncology clinical trial site shall obtain a San Francisco Health Care Facility Operating License in addition to the state authorization required under SB 1042.
- (b) The operating license shall specify the 10 categories of Physical AI systems authorized for use at the site, the maximum number of concurrent patients, and the operating schedule (24 hours per day, 7 days per week).
- (c) The license shall be renewed annually, contingent upon maintaining a cumulative PSL score within the Advanced Site band (60.0 or above) and submitting the annual safety and performance reports [9].

5.2. Inspection Requirements

- (a) The San Francisco Department of Public Health shall conduct an initial pre-operational inspection of the facility before the first patient is enrolled.

- (b) Routine inspections shall occur at least quarterly, with additional inspections triggered by adverse event reports, cybersecurity incidents, or patient complaints.
- (c) Inspections shall include verification of Physical AI system safety certifications, review of adverse event logs and resolution records, audit trail integrity checks, and observation of at least one active procedure.
- (d) Inspection findings shall be posted on the department's public website within 30 days of the inspection date.

6. Noise and Environmental Standards

- (a) The site shall comply with the San Francisco Noise Ordinance (Article 29 of the Police Code) with the following additional provisions for 24-hour Physical AI operations:
 - (1) HVAC and mechanical equipment noise at the property line shall not exceed 45 dBA during nighttime hours (10:00 PM to 7:00 AM).
 - (2) Loading and delivery activities shall be restricted to daytime hours (7:00 AM to 8:00 PM) except for emergency medical supply deliveries.
 - (3) Robot maintenance and testing activities generating noise above ambient levels shall be conducted in acoustically isolated maintenance bays.
- (b) Emergency generator testing shall be limited to daytime hours and shall not exceed one test per month lasting no more than 30 minutes.
- (c) The site shall implement a stormwater management plan compliant with San Francisco's Green Infrastructure Ordinance, incorporating the parking facility's impervious surface area.

7. Relationship to State Requirements

- (a) These municipal requirements supplement and do not replace the state-level requirements established under SB 1042, AB 2847, and SB 892.
- (b) Where a municipal requirement exceeds the corresponding state requirement, the municipal requirement shall apply. Where a state requirement exceeds the corresponding municipal requirement, the state requirement shall apply.
- (c) The three adapted regulatory frameworks (ICH E6(R3), 21 CFR Part 50, and 21 CFR Part 312) apply to Physical AI oncology clinical trial sites in San Francisco as they apply throughout the state [5, 6, 7].

California Code of Regulations, Title 22 - Physical AI Oncology Trial Site Regulations

1. General Provisions

1.1. § 75000. Authority and Purpose

- (a) These regulations are adopted pursuant to the authority granted to the California Department of Public Health (hereinafter “the department”) under the California Physical AI Oncology Clinical Trial Authorization and Site Establishment Act of 2026 (Health and Safety Code § 1399.900 et seq.).
- (b) The purpose of these regulations is to establish standards and procedures for the authorization, inspection, and oversight of Physical AI oncology clinical trial sites throughout California.
- (c) These regulations apply uniformly across the State of California. The first authorized site shall be established in San Francisco, and subsequent sites may be authorized in any California jurisdiction that meets the requirements of these regulations.

1.2. § 75001. Definitions

In addition to the definitions established in Health and Safety Code § 1399.901, the following definitions apply:

- (a) “Authorization certificate” means the document issued by the department to a qualified entity permitting the operation of a Physical AI oncology clinical trial site.
- (b) “Graduated autonomy” means the phased approach to Physical AI system deployment in which autonomy levels increase from teleoperation in Phase 1 to supervised autonomy in Phase 2 to full autonomy range in Phase 3, as established in the adapted 21 CFR Part 312 § 312.21 [7].
- (c) “Pre-procedure safety matrix” means the checklist of safety verifications that shall be completed before initiating any Physical AI-assisted clinical procedure, as defined in the adapted 21 CFR Part 50 § 50.3 [6].
- (d) “Robot capability profile” means the formal documentation of a Physical AI system’s validated task capabilities, performance envelope, and safety constraints.
- (e) “Site safety officer” means a human staff member certified by the department to supervise Physical AI system operations during a shift, with authority to initiate emergency stop procedures and override autonomous operations.

2. Authorization Process

2.1. § 75010. Application Requirements

- (a) An entity seeking authorization to operate a Physical AI oncology clinical trial site shall submit an application to the department on forms prescribed by the department, including the following:
 - (1) Identification of the applicant entity, including ownership structure, officers, and any parent organizations.
 - (2) Site location and facility description, including architectural plans, engineering specifications, and evidence of compliance with applicable building codes.
 - (3) Physical AI System Deployment Plan identifying all robot types and instances, with current USL scores for each system [4].
 - (4) Phase 0 simulation validation results for each Physical AI system type, demonstrating cross-framework validation using two or more frameworks [5].
 - (5) Staffing plan identifying all positions, qualifications, training requirements, and shift schedules for the 24-hour operating model.
 - (6) Quality management system documentation addressing Physical AI-specific processes.
 - (7) Emergency response plan addressing Physical AI system failures, cybersecurity incidents, natural disasters, and medical emergencies.
 - (8) Patient flow projections based on the simulation data showing capacity for 150 to 180 patients per day across 15 or more cancer types [1].
 - (9) Financial documentation demonstrating adequate resources to sustain operations for a minimum of 24 months.
 - (10) Evidence of compliance with adapted ICH E6(R3), 21 CFR Part 50, and 21 CFR Part 312 regulatory frameworks [5, 6, 7].
- (b) The application shall be accompanied by a nonrefundable filing fee of twenty-five thousand dollars (\$25,000).

2.2. § 75011. Review Process

- (a) The department shall assign a review team consisting of specialists in robotic engineering, oncology, clinical trial regulation, facility safety, and cybersecurity.
- (b) The review team shall complete an initial completeness review within 30 days of application receipt and notify the applicant of any deficiencies.
- (c) The department shall conduct a substantive review of the complete application and issue a decision within 120 days of receiving a complete application.

- (d) The department may conduct a pre-authorization site inspection as part of the review process.

2.3. § 75012. Authorization Certificate

- (a) An authorization certificate shall specify the following:
 - (1) The authorized Physical AI system types and maximum number of instances.
 - (2) The maximum number of concurrent patients.
 - (3) The authorized operating schedule.
 - (4) The minimum staffing requirements.
 - (5) Any conditions or limitations specific to the site.
- (b) The authorization certificate shall be valid for a period of three years, subject to annual compliance verification.
- (c) The certificate shall be prominently displayed at the site entrance and available for public inspection.

3. Physical AI System Standards

3.1. § 75020. System Registration

- (a) Each Physical AI system instance deployed at an authorized site shall be registered with the department, including the following information:
 - (1) System type, manufacturer, model, and serial number.
 - (2) Current USL score with dimension-level breakdown (simulation framework switching, AI integration, cross-robot progress sharing, clinical trial collaboration) [4].
 - (3) Current PSL scores across three regulatory dimensions (Omniscient, Omnipresent, Omnipotent) [9].
 - (4) Robot capability profile documenting validated task capabilities.
 - (5) Phase 0 simulation validation results.
 - (6) Installation date and commissioning records.
- (b) Registration shall be updated within 14 days of any change to system configuration, USL score, or capability profile.

3.2. § 75021. Minimum USL Thresholds

Physical AI systems shall meet the following minimum USL score thresholds, consistent with the MCP-PAI standard [6]:

System Category	Minimum USL Score
Surgical robots	7.0
Radiotherapy positioning robots	6.0
Robotic needle-placement systems	6.0
Radiotherapy motion-tracking robots	6.0
Imaging assistant robots	6.0
Steerable needle robots	6.0
Collaborative robots (cobots)	5.0
Rehabilitation exoskeletons	4.0
Humanoid robots	4.0
Social companion robots	3.0

3.3. § 75022. Maintenance and Calibration

- (a) Each Physical AI system shall undergo scheduled maintenance at intervals specified by the manufacturer and documented in the robot capability profile.
- (b) Maintenance windows shall be coordinated to maintain continuous clinical capability across all system categories, as demonstrated in the simulation where SURG-01 maintenance (22:15 to 01:59) was scheduled during low-demand hours with SURG-02 and SURG-03 remaining available [1].
- (c) Calibration records shall include date, calibration parameters, results, and the identity of the person or system performing the calibration.
- (d) A Physical AI system shall not be used for patient procedures until post-maintenance verification demonstrates performance within specified tolerances.

4. Staffing and Training Standards

4.1. § 75030. Minimum Staffing Requirements

- (a) An authorized Physical AI oncology clinical trial site operating on a 24-hour schedule shall maintain the following minimum staffing:
 - (1) Two to three site safety officers per shift, with three shifts covering the 24-hour operating period.
 - (2) One on-call emergency physician available within 15 minutes at all times.
 - (3) One on-call pharmacist available within 15 minutes at all times.
 - (4) One facility maintenance technician on-site per shift.
 - (5) One cybersecurity operations specialist on-call at all times.

- (b) Total full-time equivalent staffing shall be a minimum of 8 and a maximum of 10, reflecting the Physical AI model in which more patients are treated with more robots and fewer workers [3].
- (c) The site shall maintain a staffing contingency plan for situations requiring additional human personnel, such as multiple simultaneous adverse events or extended system outages.

4.2. § 75031. Training and Certification

- (a) Site safety officers shall complete a department-approved training program of not less than 160 hours covering the following:
 - (1) Physical AI system operation and capabilities for all 10 robot categories deployed at the site.
 - (2) Emergency stop procedures and manual override protocols.
 - (3) Adverse event recognition, documentation, and reporting.
 - (4) Patient interaction protocols, including patient rights under AB 2847.
 - (5) Cybersecurity awareness and incident response.
 - (6) Regulatory compliance including ICH E6(R3), 21 CFR Part 50, and 21 CFR Part 312 as adapted for Physical AI [5, 6, 7].
- (b) Certification shall be valid for two years, with 40 hours of continuing education required for renewal.
- (c) The department shall develop and maintain the training curriculum in consultation with Physical AI system manufacturers, oncology professional organizations, and patient advocacy groups.

5. Inspection and Enforcement

5.1. § 75040. Routine Inspections

- (a) The department shall conduct routine inspections of authorized sites at least quarterly, without prior notice.
- (b) Inspections shall assess compliance with authorization conditions, Physical AI system registration and maintenance, staffing requirements, patient rights protections, adverse event reporting, data protection, and facility condition.
- (c) The department shall publish inspection reports on its website within 30 days of the inspection.

5.2. § 75041. Enforcement Actions

- (a) The department may take the following enforcement actions for noncompliance:
 - (1) Written notice of deficiency with required corrective action plan.

- (2) Administrative penalty of up to ten thousand dollars (\$10,000) per day per violation.
 - (3) Suspension of authorization to use specific Physical AI system types.
 - (4) Suspension of site authorization.
 - (5) Revocation of site authorization.
- (b) Before suspending or revoking authorization, the department shall provide written notice and an opportunity for hearing in accordance with the Administrative Procedure Act (Government Code § 11500 et seq.).
- (c) In cases of imminent danger to patient safety, the department may issue an immediate suspension order, with a hearing to be held within 15 days.

6. Reporting Requirements

6.1. § 75050. Monthly Reports

- (a) Authorized sites shall submit monthly reports to the department including:
- (1) Total patients served and procedures performed, disaggregated by cancer type and Physical AI system type.
 - (2) Current PSL scores for all robot types and cumulative site PSL score.
 - (3) All adverse events with grade, resolution, and root cause analysis.
 - (4) Physical AI system utilization rates and maintenance records.
 - (5) Staffing levels and any departures from minimum requirements.

6.2. § 75051. Annual Reports

- (a) Authorized sites shall submit annual reports including all elements required under the adapted 21 CFR Part 312 § 312.33, plus the following state-specific elements [7]:
- (1) Year-over-year comparison of patient outcomes and throughput.
 - (2) Updated USL scores for all Physical AI systems with trend analysis.
 - (3) Community benefits achieved, including local workforce development.
 - (4) Patient satisfaction data and complaint resolution statistics.
 - (5) Comparison of site performance to the baseline established by the 24-hour simulation (168 patients, 99.7 percent uptime, 8-minute average wait, 4.7 out of 5.0 satisfaction) [1].

FDA Physical AI Oncology Trial Site National Compliance Guide

1. Scope and Purpose

1.1. Scope

This compliance guide identifies the federal regulations applicable to Physical AI oncology clinical trial sites and specifies where corrections, additions, and adaptations should be applied to existing regulatory text. Because a comprehensive amendment of each federal regulation is beyond the scope of a single document, this guide uses a reference-and-correction approach: for each applicable regulation, the specific sections requiring updates are identified, and the nature of the required changes is described.

1.2. Purpose

The purpose is to provide Physical AI oncology clinical trial site operators, sponsors, and regulators with a consolidated roadmap for federal compliance. This guide is informed by three detailed regulatory adaptations already completed:

- (1) Adaptation of ICH E6(R3) for Physical AI oncology clinical trials (1,301 lines, 50+ defined terms), covering good clinical practice, investigator responsibilities, sponsor responsibilities, and data governance [5].
- (2) Adaptation of 21 CFR Part 50 for Physical AI oncology clinical trials (748 lines, 18 Physical AI-specific definitions), covering human subjects protection and informed consent [6].
- (3) Adaptation of 21 CFR Part 312 for Physical AI oncology clinical trials (1,700 lines, 22 Physical AI definitions), covering IND applications, graduated autonomy, safety reporting, and decommissioning [7].

1.3. Evidence Base

The evidence for a Physical AI oncology trial transition is overwhelmingly based on the most recent simulation [1], which demonstrated:

- 168 unique patients across 15 cancer types served in 24 hours by 29 robot instances of 10 types.
- Patient throughput 10 times greater than traditional trials (150 to 180 patients per day versus 5 to 15).
- Per-patient cost reduction of 75 to 85 percent.
- Patient retention improvement from 70 to 80 percent (traditional) to 85 to 95 percent (Physical AI).

- 99.7 percent facility uptime across 24 continuous hours.
- Average patient wait time of 8 minutes.
- 7 adverse events (all Grade 1 or 2, all resolved same hour).
- Patient satisfaction of 4.7 out of 5.0.
- Cumulative site PSL of 63.4 to 64.4 (Advanced Site band).
- Both minute-level clinical detail and 24-hour operational continuity handled correctly by AI systems.
- Staffing reduced from 80 to 120 FTE to 8 to 10 FTE.
- All clinical procedures, monitoring, documentation, and regulatory compliance performed autonomously.
- Seven surgical procedures with 100 percent R0 resection rate.
- Peak utilization of 72.4 percent in hour 09 with 28 concurrent patients.

2. 21 CFR Part 50 – Protection of Human Subjects

2.1. Sections Requiring Update

The adapted 21 CFR Part 50 (DOI: 10.5281/zenodo.19040707) provides the comprehensive text [6]. The following sections of the original regulation require corrections:

- (a) **§ 50.1 (Scope):** Add Physical AI system categories and applicability criteria. Specify that the regulation applies to all clinical investigations where Physical AI systems perform, assist, or monitor procedures. Add minimum USL thresholds by robot type.
- (b) **§ 50.3 (Definitions):** Add 18 Physical AI-specific definitions including robot agent, USL, digital twin, simulation framework, MCP, hash-chained audit trail, HIPAA Safe Harbor de-identification, PCCP, federated learning, pre-procedure safety matrix, and conformance levels.
- (c) **§ 50.20 (General Requirements for Informed Consent):** Extend to require disclosure of robotic system role, autonomy mode, USL score in plain language, safety record, emergency stop provisions, digital twin usage, and MCP data processing pipeline.
- (d) **§ 50.25 (Elements of Informed Consent):** Add Physical AI elements as additional required elements, including all 10 robot categories and their patient-facing instruction formats [10].
- (e) **§ 50.50 through § 50.56 (Pediatric Protections):** Add age-appropriate robotic demonstration requirements, parental consent provisions for Physical AI procedures, and advocate appointments.

2.2. Implementation Notes

Sites should apply the full adapted text from the Physical AI 21 CFR Part 50 document [6]. Where the adapted text extends the original regulation, the extended provisions are additive and do not replace the original human subjects protection requirements.

3. 21 CFR Part 312 – Investigational New Drug Application

3.1. Sections Requiring Update

The adapted 21 CFR Part 312 (DOI: 10.5281/zenodo.19057628) provides the comprehensive text [7]. Key sections requiring corrections:

- (a) **§ 312.1-312.2 (Scope):** Extend to cover clinical investigations where Physical AI systems perform, assist, or monitor investigational drug administration. Define when IND coverage is required based on Physical AI risk introduction.
- (b) **§ 312.3 (Definitions):** Add 22 Physical AI definitions including robot agents, USL scoring (9 platforms evaluated), digital twins, simulation frameworks, robot capability profiles, task orders, emergency stop, human oversight, MCP, and VLA models.
- (c) **§ 312.21 (Phases of Investigation):** Add Phase 0 simulation validation requirements. Establish graduated autonomy: Phase 1 teleoperation only, Phase 2 graduated autonomy with safety data review, Phase 3 full autonomy range with accumulated evidence [7].
- (d) **§ 312.23 (IND Content):** Add Physical AI System Description requirements: system identification, USL score, role in drug administration, hardware and software specifications, PCCP, Phase 0 results, digital twin specifications, safety architecture, cybersecurity assessment, MCP integration, human-robot interaction protocols, and maintenance procedures.
- (e) **§ 312.30 (Protocol Amendments):** Define Physical AI-specific amendments: new robot type, autonomy level change, significant AI model update, simulation framework change, human oversight change, digital twin methodology change, and MCP architecture change.
- (f) **§ 312.32 (Safety Reporting):** Add seven Physical AI adverse event categories: serious Physical AI adverse events, Physical AI-drug interactions, cybersecurity incidents (7-day or 15-day reporting), AI model performance degradation, sim-to-real divergence (trajectory above 4mm or force above 1.0N), digital twin discrepancy, and Physical AI Safety Review Committee findings.
- (g) **§ 312.33 (Annual Reports):** Add Physical AI elements: USL score updates, AI model updates, simulation results, digital twin metrics, cybersecurity events, and safety committee findings.
- (h) **§ 312.38 (Withdrawal of IND):** Add decommissioning requirements: safe deactivation, log preservation, secure patient data erasure, 30-day decommissioning timeline, 60-day technical analysis [7].

- (i) **§ 312.40 (Requirements for Use):** Add Physical AI readiness requirements: system description submitted, USL threshold met, pre-procedure safety matrix verified, training completed, MCP infrastructure configured.
- (j) **§ 312.42 (Clinical Holds):** Add eight Physical AI-specific grounds: robotic safety failures, AI model degradation, simulation-reality divergence (above 2x tolerance), cybersecurity compromise, USL score decline, inadequate system description, digital twin failure, and inadequate human oversight.

4. ICH E6(R3) – Good Clinical Practice

4.1. Sections Requiring Update

The adapted ICH E6(R3) (DOI: 10.5281/zenodo.18973368) provides the comprehensive text [5]. Key sections requiring corrections:

- (a) **Section 1 (Principles):** Extend foundational principles to encompass robotic surgical systems, AI algorithms, and digital twins. Add Physical AI system classification (seven robot categories) and the USL scoring framework (v1.4.0 through v1.8.0) [4].
- (b) **Section 2 (Investigator Responsibilities):** Add robotic system oversight, AI model monitoring, digital twin validation, and cybersecurity incident management to investigator duties.
- (c) **Section 3 (Sponsor Responsibilities):** Add quality management for Physical AI components: robotic system selection, AI lifecycle management, simulation validation, digital twin deployment, and federated multi-site coordination.
- (d) **Section 4 (Data Governance):** Add full lifecycle governance for Physical AI data: robotic telemetry, AI decision logs, digital twin state data. Mandate hash-chained audit trails (SHA-256) and HIPAA Safe Harbor de-identification. Address five AI categories: generative AI (VLA, diffusion policies), agentic AI (LLM-based), reinforcement learning, self-supervised learning, and supervised learning.
- (e) **Appendix A (Physical AI System Documentation):** Add requirements analogous to Investigator’s Brochure for Physical AI system descriptions.
- (f) **Appendix B (Protocol Templates):** Add Physical AI-specific protocol elements including simulation validation, robot deployment plans, and graduated autonomy phases.
- (g) **Glossary:** Add 50+ terms defined including agentic AI, calibration, cobots, digital twins, federated learning, sim-to-real transfer, and USL framework.

5. 21 CFR Part 11 – Electronic Records

5.1. Sections Requiring Update

- (a) **§ 11.10 (Controls for Closed Systems):** Specify that Physical AI system telemetry, AI decision logs, and digital twin records are electronic records subject to Part 11 requirements. Hash-chained audit trails (SHA-256) satisfy the audit trail requirement when cryptographic chain integrity is verified quarterly [5].
- (b) **§ 11.30 (Controls for Open Systems):** Apply to federated learning data exchanges between Physical AI systems at different sites, requiring encryption and digital signature verification.
- (c) **§ 11.50 (Signature Manifestations):** Extend to include automated system signatures from Physical AI systems that generate, modify, or transmit electronic records during clinical procedures.

6. 21 CFR Part 820 – Quality System Regulation

6.1. Sections Requiring Update

- (a) **§ 820.30 (Design Controls):** Apply to Physical AI systems classified as medical devices, including Phase 0 simulation validation as part of design verification and validation.
- (b) **§ 820.70 (Production and Process Controls):** Apply to the deployment and configuration of Physical AI systems at trial sites, including calibration, maintenance, and software update processes.
- (c) **§ 820.90 (Nonconforming Product):** Apply to Physical AI systems that fall below minimum USL thresholds or exhibit performance degradation, with corrective action timelines aligned with the adapted 21 CFR Part 312 safety reporting requirements [7].
- (d) **§ 820.198 (Complaint Files):** Extend to include Physical AI system malfunction reports, patient complaints involving robotic systems, and AI model performance complaints.

7. Additional Federal Requirements

7.1. HIPAA (45 CFR Parts 160 and 164)

Physical AI oncology clinical trial sites shall comply with HIPAA Privacy, Security, and Breach Notification Rules with additional attention to the following Physical AI-specific considerations:

- (a) Robotic telemetry data that can be linked to specific patients constitutes protected health information (PHI) under HIPAA.
- (b) Digital twin data constitutes PHI when it contains or is derived from individually identifiable health information.

- (c) MCP-PAI server architectures shall implement the HIPAA Security Rule administrative, physical, and technical safeguards [5].
- (d) The five MCP-PAI servers (authorization, FHIR, DICOM, ledger, provenance) shall each maintain individual HIPAA compliance documentation.

7.2. Nuclear Regulatory Commission (10 CFR Part 35)

Physical AI oncology clinical trial sites operating radiotherapy systems shall comply with NRC radiation safety requirements, with the following Physical AI-specific considerations:

- (a) Radiotherapy positioning robots and motion-tracking robots shall be included in the site's radiation safety program as radiation-producing equipment accessories.
- (b) The radiation safety officer shall be trained in Physical AI system radiation safety implications.
- (c) Radiation dose calculations shall account for the precision of Physical AI positioning systems, which demonstrated submillimeter accuracy and 94.2 percent beam gating efficiency in the simulation [1].

8. Cross-Reference Matrix

The following table maps Physical AI system categories to the primary federal regulations that apply to each:

System Type	Part 50	Part 312	E6(R3)	Part 11	Part 820
Surgical robots	Yes	Yes	Yes	Yes	Yes
Cobots	Yes	Yes	Yes	Yes	Yes
RT positioning	Yes	Yes	Yes	Yes	Yes
Needle-placement	Yes	Yes	Yes	Yes	Yes
Social companion	Yes	–	Yes	Yes	–
Humanoids	Yes	–	Yes	Yes	–
RT motion-tracking	Yes	Yes	Yes	Yes	Yes
Imaging assistants	Yes	Yes	Yes	Yes	Yes
Steerable needles	Yes	Yes	Yes	Yes	Yes
Rehab exoskeletons	Yes	–	Yes	Yes	Yes

Physical AI Oncology Clinical Trial Facility Building Code Standards

1. General Requirements

1.1. Scope

These building code standards supplement the California Building Standards Code (Title 24, Parts 1 through 12) with additional requirements specific to Physical AI oncology clinical trial facilities.

These standards apply to new construction of facilities designed to house 10 categories of Physical AI robotic systems operating autonomously on a 24-hour, 7-day-per-week schedule serving up to 180 oncology patients per day [3].

1.2. Facility Size and Configuration

- (a) The minimum gross floor area shall be 85,000 square feet, with a maximum of 100,000 square feet, distributed across the clinical wings, support areas, and infrastructure spaces specified in these standards [3].
- (b) The facility shall accommodate nine clinical wings corresponding to the following Physical AI system deployment zones:
 - (1) Three surgical suites for surgical robots (SURG-01, SURG-02, SURG-03).
 - (2) Four cobot stations for collaborative biopsy robots.
 - (3) Three radiotherapy vaults for RT positioning robots.
 - (4) Two needle-placement suites for robotic needle-placement systems.
 - (5) Five social companion stations for pediatric and adult companion interactions.
 - (6) Three humanoid stations for rehabilitation engagement.
 - (7) Three RT motion-tracking vaults (co-located with RT positioning).
 - (8) Four imaging bays for imaging assistant robots.
 - (9) Two steerable needle suites.
 - (10) One rehabilitation exoskeleton bay with two active stations and one reserved station.
- (c) Support areas shall include a robot-accessible pharmacy, server room, maintenance bay, and decontamination area [1].

2. Structural Requirements

2.1. Floor Load Capacity

- (a) Surgical suites shall be designed for a minimum floor live load of 150 pounds per square foot to support surgical robot systems (including the da Vinci Xi surgical system at approximately 1,800 pounds), instrument carts, patient tables, and support equipment [10].
- (b) Radiotherapy vaults shall be designed for a minimum floor live load of 200 pounds per square foot to support linear accelerator systems, radiotherapy positioning robots, and radiation shielding components.
- (c) Imaging bays shall be designed for a minimum floor live load of 125 pounds per square foot to support imaging equipment, robotic ultrasound systems, and CT scanners used for needle-placement guidance.

- (d) Rehabilitation exoskeleton areas shall include floor-mounted parallel bars with anchorage rated for 500 pounds lateral force, supporting patients during exoskeleton-assisted gait training [10].
- (e) Server rooms shall be designed for a minimum floor live load of 250 pounds per square foot to support server rack installations for AI computation infrastructure.
- (f) General clinical circulation areas shall maintain a minimum floor live load of 100 pounds per square foot, accommodating hospital beds, wheelchairs, and mobile robot systems simultaneously.

2.2. Vibration Control

- (a) Surgical suites and needle-placement suites shall meet VC-E vibration criteria (peak velocity 3.1 micrometers per second) to support the submillimeter precision required by surgical robots and needle-placement systems [1].
- (b) Imaging bays shall meet VC-D vibration criteria (peak velocity 6.3 micrometers per second) to support diagnostic imaging quality.
- (c) Radiotherapy vaults shall meet VC-C vibration criteria (peak velocity 12.5 micrometers per second) to support accurate beam alignment.
- (d) Vibration isolation joints shall separate clinical wings from building mechanical systems, loading areas, and parking structures.

3. Mechanical Systems (HVAC)

3.1. Surgical Suite Air Handling

- (a) Surgical suites shall maintain ISO Class 7 (10,000 particles per cubic foot) or better air cleanliness using HEPA filtration with a minimum of 20 air changes per hour [3].
- (b) Surgical suites shall maintain positive pressure of at least 2.5 Pascals relative to adjacent spaces.
- (c) Temperature control shall maintain 68 to 73 degrees Fahrenheit with humidity between 30 and 60 percent relative humidity.
- (d) Independent air handling units shall serve each surgical suite to prevent cross-contamination and allow maintenance without affecting other suites.

3.2. Radiotherapy Vault Ventilation

- (a) Radiotherapy vaults shall have dedicated exhaust systems capable of removing ozone generated by high-energy radiation beams.
- (b) A minimum of 6 air changes per hour shall be maintained during treatment operations, with 10 air changes per hour during purge cycles between patients.

- (c) Exhaust air from radiotherapy vaults shall not be recirculated to other building areas.

3.3. Server Room Cooling

- (a) Server rooms supporting AI computation infrastructure shall maintain temperature between 64 and 75 degrees Fahrenheit with redundant cooling systems providing N+1 capacity [3].
- (b) Cooling systems shall reject a minimum heat load of 200 kilowatts, calculated from the server room's portion of the facility's total 800 to 1,200 kilowatt electrical load.
- (c) Server room cooling shall be on a dedicated HVAC zone independent of clinical space climate control.
- (d) Hot aisle and cold aisle containment shall be used to maximize cooling efficiency.

4. Electrical Systems

4.1. Power Supply

- (a) The facility shall be served by dual utility power feeds from separate substations or feeders, providing full redundancy for the 800 to 1,200 kilowatt total electrical load [3].
- (b) Automatic transfer switches shall provide seamless transition between utility feeds, with a maximum transfer time of 10 milliseconds for critical clinical systems.
- (c) Emergency generator capacity shall be sufficient to power all critical clinical systems (surgical suites, radiotherapy vaults, server rooms, life safety systems) for a minimum of 72 hours at full load.
- (d) Generator fuel storage shall accommodate 96 hours of operation with automatic fuel management and monitoring.

4.2. Uninterruptible Power Supply (UPS)

- (a) All surgical suites shall be served by dedicated UPS systems providing a minimum of 30 minutes of battery backup at full surgical robot operating load, sufficient to safely complete or abort any procedure in progress.
- (b) Server rooms shall be served by UPS systems providing a minimum of 15 minutes of battery backup, sufficient for orderly shutdown of AI computation systems and data preservation.
- (c) All Physical AI system charging and standby stations shall be served by UPS-backed circuits to maintain robot readiness during power transitions.
- (d) UPS systems shall be tested monthly under load, with results recorded in the facility maintenance log.

4.3. Electrical Distribution

- (a) Dedicated electrical panels shall serve each clinical wing, with sufficient circuit capacity for all Physical AI system types deployed in that wing plus 25 percent spare capacity.
- (b) Isolated ground circuits shall serve all Physical AI systems and medical imaging equipment to prevent electromagnetic interference.
- (c) Power quality monitoring systems shall continuously track voltage, current, frequency, harmonics, and transients on all circuits serving Physical AI systems, with automated alerting for out-of-specification conditions.

5. Radiation Shielding

- (a) Radiotherapy vault shielding shall comply with NCRP Report 151 and California Radiologic Health Branch requirements, designed for the maximum beam energy and workload planned for the facility.
- (b) Shielding design shall account for the Physical AI positioning robot's range of motion, ensuring that all achievable patient positions result in adequate shielding of adjacent spaces.
- (c) The primary barrier shall be designed for the full workload of the radiotherapy system, and secondary barriers shall account for leakage and scatter radiation.
- (d) Maze entrances shall be designed to reduce radiation levels at the maze entrance to below 0.02 milliSieverts per hour during beam-on conditions.
- (e) Shielding calculations shall be reviewed and certified by a qualified medical physicist before construction begins.

6. Technology Infrastructure

6.1. Network Architecture

- (a) The facility shall provide a minimum of 10 gigabits per second internal network bandwidth using redundant fiber-optic backbone infrastructure [3].
- (b) Clinical networks, administrative networks, and Physical AI system control networks shall be physically or logically segmented with firewall separation between segments.
- (c) Each clinical wing shall have dedicated network switches with redundant uplinks to the core network.
- (d) Wireless coverage (Wi-Fi 6E or later) shall be provided throughout all patient-accessible areas for patient portal access and companion robot connectivity.

6.2. Robot Charging and Docking Infrastructure

- (a) Each clinical wing shall include dedicated robot charging and docking stations rated for the specific power requirements of the Physical AI systems deployed in that wing.
- (b) Charging stations shall include automated connection systems to minimize human intervention during shift changes and standby periods.
- (c) Floor-mounted guide paths or ceiling-mounted positioning references shall facilitate autonomous robot navigation between clinical stations, charging stations, and maintenance areas.
- (d) The facility layout shall support the patient flow demonstrated in the simulation, where peak utilization reached 21 of 29 robots active simultaneously during hour 09, with an average of 8.1 of 29 active across 24 hours [1].

7. Life Safety and Egress

- (a) The facility shall comply with California Fire Code (Title 24, Part 9) and NFPA 101 Life Safety Code requirements for health care occupancies.
- (b) Emergency egress paths shall be designed to accommodate patients on hospital beds and in wheelchairs, with corridor widths of at least 8 feet in primary clinical circulation areas.
- (c) Physical AI systems shall be designed to move to designated safe positions during fire alarm activation, clearing egress paths and de-energizing non-essential systems.
- (d) Fire suppression systems in surgical suites shall use clean agent (FM-200 or Novec 1230) to protect sensitive robotic equipment while maintaining patient safety.
- (e) Fire suppression in server rooms shall use clean agent systems with pre-action activation to prevent accidental discharge damage to computing equipment.

Physical AI Oncology Clinical Trial Site Premises Code

1. Scope and Application

These premises standards govern the physical site, grounds, security systems, environmental controls, and operational conditions for Physical AI oncology clinical trial facilities authorized under California law. The standards address the unique premises requirements created by 24-hour autonomous robot operation, patient-centric on-demand scheduling, and the deployment of 29 Physical AI system instances across a 85,000 to 100,000 square-foot clinical facility [3].

The premises code complements the building code standards (structural, mechanical, electrical, technology) by addressing how the facility and its grounds are maintained, secured, and operated

during continuous clinical operations. The 24-hour simulation demonstrated that sustained operations require premises management that handles both minute-level details (individual patient movements, robot state transitions, adverse event responses) and facility-wide patterns across a full 24-hour cycle [1].

2. Site Security

2.1. Physical Security Systems

- (a) The premises shall be equipped with a comprehensive security system including the following components:
 - (1) Video surveillance covering all exterior entrances, parking areas, patient drop-off zones, lobby and reception areas, clinical wing corridors, and robot maintenance areas.
 - (2) Electronic access control on all exterior doors, all clinical wing entries, server rooms, pharmacy, robot maintenance bay, and any area housing controlled substances or sensitive equipment.
 - (3) Intrusion detection systems on all exterior doors, ground-floor windows, and roof access points.
 - (4) Duress alarm stations at the reception desk, each shift supervisor station, and the security monitoring room.
- (b) Security systems shall operate continuously during the 24-hour operating period without interruption.
- (c) Security camera recordings shall be retained for a minimum of 90 days and made available to the department and law enforcement upon request.

2.2. Access Control Zones

The premises shall be divided into the following access control zones:

- (a) **Public Zone:** Lobby, reception area, patient waiting areas, restrooms, and the covered patient drop-off canopy. Accessible to patients, visitors, and the public during operating hours.
- (b) **Patient Zone:** Clinical wings, examination rooms, procedure suites, recovery areas, and patient circulation corridors. Accessible to enrolled patients with valid credentials and accompanying persons.
- (c) **Clinical Zone:** Areas where Physical AI systems perform clinical procedures, including surgical suites, RT vaults, needle-placement suites, and steerable needle suites. Restricted to authorized clinical personnel, site safety officers, and patients undergoing procedures.
- (d) **Technical Zone:** Server rooms, robot maintenance bay, decontamination area, and equipment storage. Restricted to authorized technical personnel and site safety officers.

- (e) **Administrative Zone:** Staff offices, break rooms, training rooms, and supply storage. Restricted to site personnel.

2.3. Cybersecurity of Physical Premises

- (a) Physical AI system control networks shall be accessible only from within the Technical Zone, with no wireless access points connected to robot control systems.
- (b) The MCP-PAI server infrastructure shall be housed in the Technical Zone server room with physical access limited to authorized cybersecurity and system administration personnel [5].
- (c) Network infrastructure components (switches, routers, fiber distribution panels) shall be housed in locked enclosures within the Technical Zone.
- (d) Visitor and patient wireless networks shall be physically isolated from clinical and Physical AI system networks.

3. Patient Access and Flow

3.1. Arrival and Reception

- (a) The main patient entrance shall be accessible 24 hours per day, 7 days per week, consistent with the on-demand scheduling model that demonstrated patient arrivals across all 24 hours of the simulation, including 15 arrivals during the peak hour (hour 09) and continued arrivals during overnight hours [1].
- (b) A covered patient drop-off zone shall accommodate a minimum of three vehicles simultaneously, with a covered walkway extending at least 20 feet to the building entrance.
- (c) The reception area shall include automated check-in kiosks that verify patient identity, confirm scheduled procedures, and direct patients to the appropriate clinical wing.
- (d) The reception area shall include seating for a minimum of 20 patients and accompanying persons, reflecting the simulation's peak concurrent on-site population of 28 patients [1].

3.2. Internal Patient Circulation

- (a) Primary patient circulation corridors shall be a minimum of 8 feet wide to accommodate hospital beds, wheelchairs, rehabilitation exoskeletons, and mobile Physical AI systems simultaneously.
- (b) Wayfinding shall use a combination of visual signage, floor markings, digital displays, and optional auditory guidance compatible with the 10 Physical AI system categories that patients may encounter [10].
- (c) Patient flow through the facility shall follow the patterns validated in the simulation, with separate intake, procedure, and recovery pathways to minimize cross-traffic and wait times [1].

- (d) Recovery areas adjacent to each procedure zone shall accommodate patients during the post-procedure observation periods specified in the patient robot instructions (ranging from 2 minutes for imaging to 60 minutes for steerable needle and surgical procedures) [10].

4. Environmental Standards

4.1. Exterior Grounds

- (a) The site shall maintain landscaping and hardscaping consistent with the character of the surrounding neighborhood and the San Francisco Planning Code requirements for the applicable zoning district.
- (b) Exterior lighting shall provide a minimum of 5 foot-candles at all pedestrian pathways, parking areas, and building entrances, ensuring patient safety during the 24-hour operating period.
- (c) A minimum of 10 percent of the total lot area shall be dedicated to landscaped open space, which may include patient garden areas accessible from the Public Zone.
- (d) Signage shall clearly identify the facility as a Physical AI oncology clinical trial site, with the authorization certificate number displayed at the main entrance.

4.2. Interior Environment

- (a) Patient waiting areas and recovery spaces shall maintain interior temperatures between 68 and 74 degrees Fahrenheit with ambient noise levels not exceeding 45 dBA.
- (b) Lighting in patient areas shall be adjustable between 200 and 500 lux to accommodate different times of day and patient comfort needs during 24-hour operations.
- (c) Clinical procedure areas shall maintain lighting appropriate to the specific Physical AI system requirements, ranging from high-intensity surgical lighting in surgical suites to reduced ambient lighting in imaging bays and RT vaults.
- (d) Air quality in all patient-accessible areas shall be maintained at a minimum of MERV 13 filtration with continuous monitoring of CO₂, temperature, and humidity.

5. Robot Operational Zones

5.1. Robot Movement Corridors

- (a) Designated robot movement corridors shall connect clinical wings to the robot maintenance bay, charging stations, and decontamination area.
- (b) Robot corridors shall be separated from primary patient circulation pathways where possible, using time-of-day scheduling where physical separation is not feasible.
- (c) Floor surfaces in robot corridors shall be smooth, level (maximum slope 1:48), and free of thresholds or transitions that could impede autonomous robot navigation.

- (d) Robot corridors shall be equipped with proximity sensors and visual warning indicators (flashing lights) to alert human occupants when robots are in transit.

5.2. Robot Maintenance Bay

- (a) The robot maintenance bay shall accommodate a minimum of three Physical AI systems simultaneously for scheduled maintenance, calibration, or repair.
- (b) The bay shall include dedicated workstations with electrical, compressed air, and network connections appropriate for each Physical AI system category.
- (c) Spare parts storage shall be maintained for critical components of all 10 Physical AI system types, with inventory levels sufficient to support the maintenance schedule demonstrated in the simulation (where SURG-01 maintenance was conducted from 22:15 to 01:59 without disrupting clinical operations) [1].
- (d) The maintenance bay shall be acoustically isolated from patient areas to allow maintenance activities during nighttime hours without disturbing patients in adjacent recovery areas.

6. Emergency Access and Evacuation

- (a) Emergency vehicle access shall be provided from at least two directions, with a minimum 26-foot-wide fire access lane along the full length of the building.
- (b) A dedicated ambulance bay shall accommodate a minimum of two ambulances simultaneously, separate from the patient drop-off zone.
- (c) Evacuation plans shall account for the specific needs of oncology patients who may be connected to treatment equipment, recovering from procedures, or using rehabilitation exoskeletons at the time of evacuation.
- (d) Physical AI systems shall execute automated safe-shutdown and position procedures upon receiving evacuation signals, including retracting surgical arms, lowering treatment couches, and moving to designated safe positions to clear evacuation routes [10].
- (e) Evacuation drills shall be conducted quarterly, including scenarios specific to Physical AI system emergencies such as robot control system failure, surgical suite power loss during a procedure, and cybersecurity incidents requiring physical network isolation.

7. Waste Management

- (a) Medical waste generated by Physical AI-assisted oncology procedures shall be managed in accordance with the California Medical Waste Management Act (Health and Safety Code § 117600 et seq.).
- (b) Radioactive waste from radiotherapy operations shall be managed in accordance with applicable NRC and California Radiologic Health Branch requirements.

- (c) Robot decontamination waste shall be classified and managed based on the type of contamination (biological, chemical, or radioactive).
- (d) Dedicated waste staging areas shall be located in the Technical Zone with separate containers for biohazardous, radioactive, pharmaceutical, and general waste.
- (e) Waste collection shall be scheduled to avoid conflicts with patient arrival and departure patterns, using the patient flow data from the simulation to identify optimal collection windows [1].

Physical AI Oncology Clinical Trial Site

Parking and Patient Transportation Standards

1. Scope and Purpose

These standards govern the design, construction, and operation of dedicated parking facilities and patient transportation infrastructure for Physical AI oncology clinical trial sites authorized in California. The parking facility is a new-construction structure purpose-built to serve the trial site, designed to accommodate the unique patient flow patterns of a 24-hour, on-demand oncology clinical trial.

The 24-hour simulation demonstrated arrival patterns ranging from 2 patients during overnight hours (hours 02 through 04) to 15 patients during peak hour (hour 09), with a total of 168 unique patients in 24 hours and a peak concurrent on-site population of 28 patients [1]. The parking facility must accommodate these variable demand patterns while maintaining the 8-minute average wait time and the 24/7 accessibility that define the Physical AI trial format [2].

2. Parking Facility Design

2.1. Capacity Requirements

- (a) The parking facility shall provide a minimum of 120 parking spaces, calculated as follows:
 - (1) 80 patient and visitor spaces, based on the peak concurrent on-site population of 28 patients plus accompanying persons and a turnover factor reflecting variable procedure durations (15 minutes for imaging to 180 minutes for surgery) [1, 10].
 - (2) 15 staff spaces for the 8 to 10 full-time equivalent staff plus on-call personnel across three shifts [3].
 - (3) 10 accessible spaces (8.3 percent of total), exceeding the minimum ADA and California Building Code Chapter 11B requirements for medical facilities.
 - (4) 5 short-term patient drop-off and pick-up spaces adjacent to the building entrance.
 - (5) 10 spaces designated for ride-share and medical transport vehicle staging.

- (b) The facility shall accommodate future expansion of up to 50 percent additional capacity through structural design that supports the addition of one parking level.

2.2. Structure and Configuration

- (a) The parking facility shall be a new-construction structure, which may be either a surface lot or a multi-level structure (not to exceed three levels above grade), depending on the site constraints and zoning requirements.
- (b) If a multi-level structure is constructed, all patient parking levels shall be served by a minimum of two elevators and one ramp system meeting ADA accessibility standards.
- (c) The facility shall provide a covered pedestrian connection from the parking structure to the clinical building entrance, with a maximum walking distance of 300 feet from the most distant parking space to the building entrance.
- (d) Clear ceiling height shall be a minimum of 8 feet 2 inches on all levels to accommodate medical transport vehicles and passenger vans.
- (e) Drive aisles shall be a minimum of 24 feet wide for two-way traffic, with parking spaces a minimum of 9 feet wide by 18 feet deep.

2.3. Accessible Parking

- (a) Accessible parking spaces shall be located at the level closest to the building entrance, with a maximum distance of 100 feet from the space to the building entry.
- (b) A minimum of two van-accessible spaces (8 feet wide with 8-foot access aisle) shall be provided.
- (c) Accessible routes from parking to the building entrance shall be covered, illuminated to a minimum of 10 foot-candles, and free of steps, steep grades (maximum slope 1:20), or other barriers.
- (d) Accessible parking signage shall include the international symbol of accessibility and shall note that the facility operates 24 hours per day for on-demand oncology care.

3. Patient Drop-Off Zone

- (a) A dedicated patient drop-off and pick-up zone shall be located immediately adjacent to the main building entrance, separated from the parking facility access drive.
- (b) The drop-off zone shall accommodate a minimum of three vehicles simultaneously under a covered canopy extending at least 20 feet from the building facade.
- (c) The drop-off zone shall include a level, non-slip surface, curb cuts, and seating for patients waiting for transportation.
- (d) A call button or intercom shall allow patients to request assistance from site safety officers for transfer from vehicles to the building entrance.

- (e) The drop-off zone shall be operational 24 hours per day, with adequate lighting (minimum 10 foot-candles) and weather protection for all hours of operation, consistent with the simulation's demonstrated overnight patient arrivals and departures [1].

4. Transportation Demand Management

4.1. Public Transit Integration

- (a) The site shall be located within one-quarter mile of a BART station or Muni stop, or the site operator shall provide a shuttle service connecting the site to the nearest transit hub at intervals not exceeding 30 minutes during all operating hours.
- (b) Real-time transit information displays shall be installed in the patient waiting area and the parking facility lobby, showing schedules for BART, Muni, and site shuttle services.
- (c) The site shall participate in the Clipper card program or equivalent regional transit payment system to facilitate patient transit use.

4.2. Ride-Share and Medical Transport

- (a) The site shall designate a minimum of 10 parking spaces for ride-share and medical transport vehicle staging, located near the patient drop-off zone but not blocking drop-off lane traffic flow.
- (b) The site shall coordinate with ride-share services to establish a designated pick-up and drop-off geofence that directs drivers to the correct facility entrance.
- (c) For patients who require medical transport following procedures (such as those recovering from anesthesia after surgical robot procedures lasting 90 to 180 minutes), the site shall coordinate discharge scheduling with transport providers to minimize patient wait times [10].

4.3. Bicycle and Pedestrian Access

- (a) The site shall provide a minimum of 20 Class I (enclosed and secured) bicycle parking spaces and 10 Class II (outdoor rack) bicycle parking spaces.
- (b) Bicycle parking shall be located within 100 feet of the building entrance and shall include electric bicycle charging capability for at least 4 spaces.
- (c) Pedestrian access from the public sidewalk to the building entrance shall be clearly marked, well-lit (minimum 5 foot-candles), and separated from vehicle traffic.

5. 24-Hour Parking Operations

5.1. Staffing and Monitoring

- (a) The parking facility shall be monitored 24 hours per day through video surveillance and emergency call stations, consistent with the 24-hour operating schedule of the clinical facility.
- (b) Emergency call stations shall be located at each elevator lobby, at each stairwell entrance, and at intervals not exceeding 200 feet in the parking area.
- (c) A parking management system shall track occupancy in real time and display available spaces at the facility entrance, directing arriving patients to available spaces efficiently.
- (d) During overnight hours (10:00 PM to 6:00 AM), the parking facility may reduce lighting to 75 percent of daytime levels in unoccupied areas while maintaining full lighting in occupied areas, pedestrian routes, and the patient drop-off zone.

5.2. Electric Vehicle Support

- (a) A minimum of 20 percent of parking spaces (24 spaces) shall be equipped with Level 2 electric vehicle charging stations (240V, 32A minimum).
- (b) A minimum of 4 parking spaces shall be equipped with Level 3 DC fast charging stations (minimum 50 kW), enabling patients to charge during shorter procedures such as imaging (10 to 20 minutes) or companion robot sessions (10 to 20 minutes) [10].
- (c) An additional 20 percent of parking spaces (24 spaces) shall be designed as EV-capable, with conduit and panel capacity installed during construction to support future charging station installation.
- (d) EV charging shall be available 24 hours per day, consistent with the facility's operating schedule.

6. Emergency Vehicle Access

- (a) A dedicated ambulance bay shall be provided separate from both the parking facility and the patient drop-off zone, accommodating a minimum of two ambulances simultaneously.
- (b) Fire department access lanes shall be maintained along the full perimeter of both the clinical building and the parking facility, with a minimum width of 26 feet and a minimum vertical clearance of 13 feet 6 inches.
- (c) Emergency vehicle access routes shall be clearly marked and kept unobstructed at all times, with automatic enforcement through surveillance and notification systems.
- (d) The parking facility design shall not impede emergency vehicle access to the clinical building from any direction.

7. Sustainability

- (a) The parking facility shall include a stormwater management system compliant with the San Francisco Green Infrastructure Ordinance, incorporating bioretention areas, permeable pavement, or equivalent green infrastructure techniques.
- (b) Rooftop areas of the parking structure, if applicable, shall incorporate either solar photovoltaic panels or a green roof system covering a minimum of 50 percent of the roof area.
- (c) Parking facility lighting shall use LED fixtures with occupancy sensors and daylight harvesting to minimize energy consumption during the 24-hour operating period.
- (d) Construction materials shall meet the CalGreen (Title 24, Part 11) requirements for recycled content and low-emission materials.

Physical AI Oncology Clinical Trial Site Activation and Standard Operating Procedures

1. Site Activation Checklist

1.1. Pre-Activation Requirements

The following requirements shall be completed before the first patient is enrolled at a Physical AI oncology clinical trial site:

(a) **Regulatory Approvals:**

- (1) State authorization certificate issued by CDPH (per SB 1042).
- (2) San Francisco conditional use permit and health care facility operating license (or equivalent local permits for non-San Francisco sites).
- (3) IRB approval for the Physical AI oncology trial protocol.
- (4) IND clearance with Physical AI System Description accepted by FDA [7].
- (5) All Physical AI systems registered with CDPH with current USL scores meeting minimum thresholds [4].

(b) **Facility Readiness:**

- (1) Building occupancy permit issued.
- (2) All Physical AI systems installed, commissioned, and verified.
- (3) Phase 0 simulation validation completed for all system types with cross-framework consistency documented [5].

- (4) Network infrastructure operational with MCP-PAI five-server topology configured to the required conformance level.
- (5) Power systems tested including UPS and generator transfer.
- (6) HVAC systems verified for all clinical zones.
- (7) Security systems operational with access control zones configured.
- (c) **Staffing Readiness:**
 - (1) All site safety officers trained and CDPH-certified.
 - (2) On-call emergency physician and pharmacist agreements executed.
 - (3) Maintenance technician assigned and trained.
 - (4) Cybersecurity operations specialist on-call roster established.
 - (5) All staff completed competency assessments for Physical AI systems.
- (d) **Documentation:**
 - (1) Standard operating procedures finalized for all 10 robot types.
 - (2) Patient-facing robot instructions available in English and the most common languages spoken in the service area [10].
 - (3) Informed consent documents approved by IRB with Physical AI disclosures [6].
 - (4) Emergency response plan posted and distributed.
 - (5) Adverse event reporting workflows tested.

2. Shift Management

2.1. Three-Shift Structure

- (a) The site shall operate on a three-shift schedule covering 24 continuous hours:
 - (1) Day shift: 06:00 to 14:00.
 - (2) Evening shift: 14:00 to 22:00.
 - (3) Night shift: 22:00 to 06:00.
- (b) Each shift shall be staffed by a minimum of two site safety officers, with a third officer added during peak demand periods (typically 08:00 to 16:00 based on the simulation's patient arrival patterns showing 15 arrivals during hour 09) [1].
- (c) Shift handoff procedures shall include a structured briefing covering:
 - (1) Current patient census and active procedures.
 - (2) Physical AI system status (active, standby, maintenance) for all 29 instances.
 - (3) Any open adverse events or safety concerns.

- (4) Scheduled maintenance windows (such as the SURG-01 22:15 to 01:59 maintenance demonstrated in the simulation) [1].
- (5) Pending patient arrivals and departures.

2.2. Shift Performance Metrics

Each shift shall track and report the following metrics, consistent with the hourly data structure validated in the 24-hour simulation:

- (a) Patient arrivals, procedures completed, and departures during the shift.
- (b) Physical AI system utilization rates per type (the simulation averaged 8.1 of 29 active, with peak of 21 of 29 active) [1].
- (c) Average patient wait time (target: 8 minutes or less).
- (d) Adverse events with grade, resolution time, and root cause.
- (e) PSL score changes per robot type per regulatory dimension (maximum 0.3 change per hour per dimension) [9].
- (f) Any emergency stop activations or manual overrides.

3. Patient Processing Workflow

3.1. Arrival to Procedure

- (a) Patient arrival and check-in via automated kiosk (estimated 2 minutes).
- (b) Identity verification and informed consent confirmation. Physical AI disclosures reviewed for any system changes since enrollment.
- (c) Pre-procedure safety matrix completed, including verification that all assigned Physical AI systems meet USL thresholds and have passed daily calibration checks [6].
- (d) Patient escorted or directed to the appropriate clinical wing based on scheduled procedure type.
- (e) Patient-facing robot instructions reviewed with the patient for the specific systems involved in their procedure. For example, surgical robot patients receive the three-step surgical robot instruction sheet covering anesthesia induction, robot-assisted procedure, and post-operative recovery [10].
- (f) Procedure initiated under Physical AI system autonomous control with site safety officer oversight.

3.2. Procedure to Departure

- (a) Procedure completed; Physical AI system performs automated post-procedure protocols (retracting instruments, lowering couches, generating procedure reports).

- (b) Patient transferred to recovery area; recovery duration determined by procedure type (2 minutes for imaging to 60 minutes for surgical or steerable needle procedures) [10].
- (c) Post-procedure vital sign monitoring by Physical AI systems until discharge criteria met.
- (d) Discharge documentation generated automatically; patient receives procedure summary and follow-up instructions.
- (e) Patient exits via reception area; transportation coordination if needed.

4. Physical AI System Daily Operations

4.1. Daily System Checks

- (a) Each Physical AI system shall undergo an automated daily verification check before the first patient procedure, including:
 - (1) Mechanical calibration verification (positioning accuracy, force limits, range of motion).
 - (2) Software system integrity check (AI model version, parameter verification, hash validation).
 - (3) Network connectivity and MCP-PAI server communication test.
 - (4) Emergency stop functionality test.
 - (5) Sensor calibration verification (cameras, force sensors, proximity sensors).
- (b) Systems that fail any daily verification check shall be placed in maintenance status and shall not be used for patient procedures until the issue is resolved and the check is passed.
- (c) Daily check results shall be recorded in the system's audit trail with timestamp and pass or fail status for each verification element.

4.2. System Utilization Management

- (a) The site's scheduling system shall manage Physical AI system utilization to balance patient demand against system availability, following the utilization patterns validated in the simulation:
 - (1) RT positioning robots: 38 percent average utilization.
 - (2) Imaging assistants: 38 percent average utilization.
 - (3) RT motion-tracking robots: 37 percent average utilization.
 - (4) Surgical robots: scheduled with appropriate intervals between procedures for decontamination and preparation.
 - (5) Companion robots and humanoids: flexible scheduling to accommodate patient emotional support needs [1].

- (b) When utilization approaches 80 percent for any system type, the scheduling system shall extend appointment intervals to prevent queue buildup and maintain the target 8-minute average wait time.
- (c) Maintenance windows shall be scheduled during periods of historically low demand (typically 22:00 to 06:00), using the simulation data as the baseline demand model [1].

5. Quality Assurance

5.1. Continuous Quality Monitoring

- (a) The site shall maintain a quality management system compliant with the adapted ICH E6(R3) requirements for Physical AI trials [5].
- (b) Key quality indicators shall be tracked continuously:
 - (1) Patient throughput versus the 168-patient 24-hour baseline.
 - (2) Adverse event rate versus the 4.2 percent baseline (7 events across 168 patients) [1].
 - (3) Same-hour adverse event resolution rate versus the 100 percent baseline.
 - (4) Fleet uptime versus the 99.7 percent baseline.
 - (5) Patient satisfaction versus the 4.7 out of 5.0 baseline.
 - (6) Cumulative PSL score versus the 63.4 to 64.4 baseline range.
- (c) Deviations from baseline quality indicators shall trigger root cause analysis and corrective action within 7 days.

5.2. Regulatory Compliance Monitoring

- (a) Compliance with the three adapted regulatory frameworks shall be monitored continuously:
 - (1) ICH E6(R3): GCP compliance for Physical AI systems, data governance, audit trail integrity [5].
 - (2) 21 CFR Part 50: Informed consent completeness, patient rights adherence, pediatric protections [6].
 - (3) 21 CFR Part 312: IND compliance, safety reporting timelines, annual report preparation [7].
- (b) Monthly compliance reports shall be submitted to CDPH as required by the state regulations.
- (c) The Physical AI Safety Review Committee shall meet at intervals not exceeding 90 days to review all safety data, quality metrics, and compliance status [7].

6. Adverse Event Management

- (a) Adverse event detection, documentation, and initial response shall be automated by Physical AI systems, with site safety officer notification within 1 minute of detection.
- (b) The seven adverse event categories defined in the adapted 21 CFR Part 312 shall be used for classification: serious Physical AI adverse events, Physical AI-drug interactions, cybersecurity incidents, AI model degradation, sim-to-real divergence, digital twin discrepancy, and safety committee findings [7].
- (c) The site shall target same-hour resolution for all Grade 1 and Grade 2 events, matching the simulation performance where all seven adverse events (nausea Grade 1, hypotension Grade 1, bleeding Grade 1, pain Grade 2, cough Grade 1, desaturation Grade 1, anxiety Grade 1) were resolved within the hour of occurrence [1].
- (d) Grade 3 or higher events shall trigger immediate escalation to the on-call emergency physician and the Physical AI Safety Review Committee.
- (e) All adverse events shall be reported to CDPH and FDA within the timelines specified in the adapted 21 CFR Part 312 § 312.32 (7 days for fatal or life-threatening, 15 days for serious) [7].

Physical AI Oncology Clinical Trial Site Emergency Preparedness Plan

1. Emergency Classification System

Physical AI oncology clinical trial sites shall classify emergencies into four levels, each with defined response protocols:

1.1. Level 1 – Minor Incident

- (a) Definition: An event affecting a single Physical AI system or a single patient without immediate risk of serious harm. Examples include a single robot entering fault mode, a Grade 1 adverse event, or a minor network disruption affecting non-critical systems.
- (b) Response: The affected Physical AI system is automatically placed in safe mode. The site safety officer assesses the situation and determines whether the system can be restarted or requires maintenance. Patient care continues using alternative systems. The simulation demonstrated that the fleet of 29 robot instances provides redundancy within each system category, allowing continued operations when individual systems are unavailable [1].
- (c) Reporting: Documented in the shift log and included in the monthly report to CDPH.

1.2. Level 2 – Moderate Emergency

- (a) Definition: An event affecting multiple Physical AI systems, multiple patients, or requiring activation of emergency medical response. Examples include multiple simultaneous robot faults, a Grade 2 or Grade 3 adverse event, a localized power failure, or a cybersecurity anomaly.
- (b) Response: The site safety officer activates the emergency response team. The on-call emergency physician is notified. Affected clinical areas are secured. Patient procedures in unaffected areas may continue. All active procedures in affected areas are safely concluded or aborted using the emergency stop protocols described in the patient robot instructions [10].
- (c) Reporting: Reported to CDPH within 72 hours. If a cybersecurity incident with potential patient harm, reported within 7 days per the adapted 21 CFR Part 312 § 312.32 [7].

1.3. Level 3 – Major Emergency

- (a) Definition: An event requiring partial or complete facility evacuation, affecting the majority of Physical AI systems, or involving a serious adverse event (Grade 4 or higher). Examples include building-wide power failure exceeding UPS capacity, structural damage, confirmed cybersecurity breach affecting patient safety, or a mass casualty event.
- (b) Response: The site administrator activates the facility emergency plan. All Physical AI systems execute automated safe-shutdown procedures. Patient evacuation begins per the evacuation plan. External emergency services are activated. The Physical AI Safety Review Committee is convened within 24 hours [7].
- (c) Reporting: Reported to CDPH and FDA immediately. Fatal or life-threatening events reported within 7 days per § 312.32. Serious events reported within 15 days [7].

1.4. Level 4 – Catastrophic Event

- (a) Definition: An event rendering the facility completely non-operational, such as a major earthquake, widespread infrastructure failure, or complete cybersecurity compromise.
- (b) Response: Complete facility evacuation. All Physical AI systems de-energized. Business continuity plan activated. Patient care transferred to partner facilities. IND sponsor notified for potential clinical hold assessment [7].
- (c) Reporting: Immediate notification to CDPH, FDA, and local emergency management. Full incident report within 30 days.

2. Physical AI System Emergency Procedures

2.1. Emergency Stop Protocol

- (a) Every Physical AI system shall have a clearly marked, accessible emergency stop mechanism that can be activated by site safety officers, clinical personnel, or patients where clinically appropriate.
- (b) Upon emergency stop activation, the system shall:
 - (1) Immediately cease all autonomous motion.
 - (2) Retract any instruments or probes that are in contact with or inside the patient, using a controlled withdrawal sequence.
 - (3) Lower treatment couches to the lowest safe position.
 - (4) De-energize radiation sources (RT systems).
 - (5) Maintain essential monitoring and life safety functions.
 - (6) Generate an emergency stop event record in the audit trail with timestamp, trigger source, and system state at the time of activation [5].
- (c) The patient robot instructions for each system type include specific emergency behavior; for example, surgical robots retract arms through chest ports, RT positioning systems lower couches to exit height, and exoskeletons return to standing rest position [10].

2.2. Cybersecurity Incident Response

- (a) The cybersecurity incident response plan shall address Physical AI-specific threats:
 - (1) Robot control system compromise (unauthorized commands to Physical AI systems).
 - (2) AI model poisoning (manipulation of AI model parameters or training data).
 - (3) Digital twin data manipulation (alteration of patient-specific models).
 - (4) MCP-PAI server infrastructure attack (compromise of authorization, FHIR, DICOM, ledger, or provenance servers).
 - (5) Network segmentation breach (traffic between isolated network zones).
- (b) Upon detection of a cybersecurity incident:
 - (1) All affected Physical AI systems are immediately isolated from the network.
 - (2) Active procedures on affected systems are safely concluded or aborted.
 - (3) The cybersecurity operations specialist assesses the scope and severity.
 - (4) Patient data integrity is verified using hash-chained audit trail validation [5].
 - (5) The incident is classified and reported per the adapted 21 CFR Part 312 timelines [7].

- (c) The site shall conduct annual penetration testing and quarterly vulnerability assessments of all Physical AI system interfaces and MCP-PAI infrastructure.

3. Natural Disaster Preparedness

3.1. Seismic Preparedness

- (a) The facility shall be designed to Seismic Design Category D or higher per ASCE 7 and the California Building Code, appropriate for San Francisco's seismic zone.
- (b) Physical AI systems shall be seismically anchored to their mounting surfaces with anchorage rated for the systems' operating weight plus 50 percent margin.
- (c) Automated seismic sensors shall trigger Physical AI system safe-mode procedures when ground acceleration exceeds 0.1g, including retracting surgical instruments, lowering couches, and pausing radiation delivery.
- (d) Post-earthquake inspection protocols shall verify Physical AI system calibration before resuming patient procedures.

3.2. Power Failure

- (a) The dual utility feed design provides primary power redundancy. If both feeds fail, the automatic transfer switch engages emergency generators within 10 seconds.
- (b) UPS systems bridge the transfer gap, providing 30 minutes for surgical suites (sufficient to complete or safely abort any procedure) and 15 minutes for server rooms (sufficient for orderly AI system shutdown and data preservation).
- (c) If generator fuel is projected to be exhausted before utility power restoration, the site shall begin controlled patient discharge and Physical AI system shutdown, prioritizing completion of active procedures.
- (d) The 72-hour generator fuel capacity provides sufficient time for coordinated patient transfer to partner facilities in extended outage scenarios.

4. Clinical Emergency Procedures

4.1. Medical Emergency During Physical AI Procedure

- (a) If a patient experiences a medical emergency during a Physical AI-assisted procedure:
 - (1) The Physical AI system's vital sign monitoring detects the emergency and initiates automated response protocols.
 - (2) Emergency stop is activated automatically or by the site safety officer.
 - (3) The on-call emergency physician is notified immediately.

- (4) The Physical AI system retracts instruments and positions the patient for emergency access.
 - (5) Resuscitation equipment is deployed from the emergency station located within each clinical wing.
 - (6) The event is documented in real time through the Physical AI system's audit trail and the patient's electronic health record.
- (b) The simulation demonstrated that real-time vital sign monitoring at minute-level resolution enables early detection of adverse events; all seven adverse events in the simulation were detected, graded, and managed through the automated monitoring system [1].

4.2. Multiple Simultaneous Adverse Events

- (a) If multiple adverse events occur simultaneously (which did not occur in the simulation but is planned for), the site shall:
- (1) Triage events by severity grade, prioritizing the highest-grade event.
 - (2) Activate the staffing contingency plan to bring additional personnel on-site within 30 minutes.
 - (3) Suspend new patient intake until the simultaneous events are resolved.
 - (4) Notify the Physical AI Safety Review Committee if the events suggest a systemic issue affecting multiple Physical AI systems or patients.

5. Business Continuity

- (a) The site shall maintain mutual aid agreements with at least two hospital or clinical facilities within 30 minutes travel time for patient transfer during extended emergencies.
- (b) Critical Physical AI system configuration data, patient records, and audit trails shall be backed up to a geographically remote data center with recovery point objective (RPO) of 15 minutes and recovery time objective (RTO) of 4 hours.
- (c) The business continuity plan shall include provisions for resuming Physical AI operations following a major emergency, including system recalibration, Phase 0 re-validation if systems were physically damaged, and regulatory notification of operational restart.
- (d) Annual business continuity exercises shall test the full emergency response chain from emergency detection through patient transfer, Physical AI system shutdown, data preservation, and operational restart.

6. Training and Drills

- (a) All site personnel shall complete emergency preparedness training as part of the 160-hour site safety officer certification program.
- (b) Emergency drills shall be conducted quarterly, rotating through the following scenarios:
 - (1) Quarter 1: Physical AI system failure during active procedure (emergency stop, patient safety, system recovery).
 - (2) Quarter 2: Cybersecurity incident (network isolation, data integrity verification, incident reporting).
 - (3) Quarter 3: Building evacuation (Physical AI safe-shutdown, patient evacuation, external emergency services coordination).
 - (4) Quarter 4: Extended power failure (generator operation, controlled patient discharge, business continuity activation).
- (c) Drill performance shall be documented and reviewed by the Physical AI Safety Review Committee, with corrective actions for any deficiencies identified.
- (d) The patient journey demonstrated that 10 distinct stages of care, from pre-screening through data lock, require coordinated emergency response capabilities at each stage [8].

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